FR30xx-C 硬件应用指南

Dual mode Bluetooth Hardware Application Guide

Rev：V1.4.5

2025.09.11

www.freqchip.com

FreqChip Confi

FREQCHIP
富芮坤

目 录

1. FR30xx-C 系列芯片简介 1
2. 封装定义 2
   2.1 FR3066DQ-C 封装 2
   2.2 FR3091E-C 封装 3
   2.3 FR5092E-C 封装 4
   2.4 FR3068E-C 封装 5
   2.5 FR30xx-C 引脚定义 6
   2.6 FR3066DQ-C 系列 GPIO 功能复用表 16
   2.7 FR3091E-C/ FR5092E-C 系列 GPIO 功能复用表 18
   2.8 FR3068E-C 系列 GPIO 功能复用表 21
3. 参考设计 24
   3.1 FR3066DQ-C 参考设计 24
   3.2 FR3091E-C 参考设计 25
   3.3 FR5092E-C 参考设计 26
   3.4 FR3068E-C 参考设计 27
4. 应用设计注意事项 28
   4.1 电源 28
   4.2 射频 29
   4.3 晶振 30
   4.4 GPIO 31
   4.5 系统设计 31
   4.6 应用接口 32
      4.6.1 USB 32
      4.6.2 I2C 32
      4.6.3 I2S 33
      4.6.4 SD 33
      4.6.5 外接 2.4G PA 34
5. PCB 天线 35
6. 回流焊 36

联系方式 Contact Information 37

修订历史 Revision History 37

FreqChip Confidential

1. FR30xx-C 系列芯片简介

- FR30xx-C 集成 RF、Baseband、PMU 等模块，支持蓝牙 V5.3(BR / EDR / LE)协议。适用于双模蓝牙 HCI 控制器，高安全性、可靠性的蓝牙 MCU 等应用。
- 布局合理，走线清晰，接地良好是做好设计的基本要求。FR30xx-C 较少的外围器件，可以使用双面板进行设计，从而节省成本。双面板设计时 PCB 顶层用于摆件和走信号线，底层走电源线并尽可能保证一个完整的地平面。
- FR30xx-C 系列型号

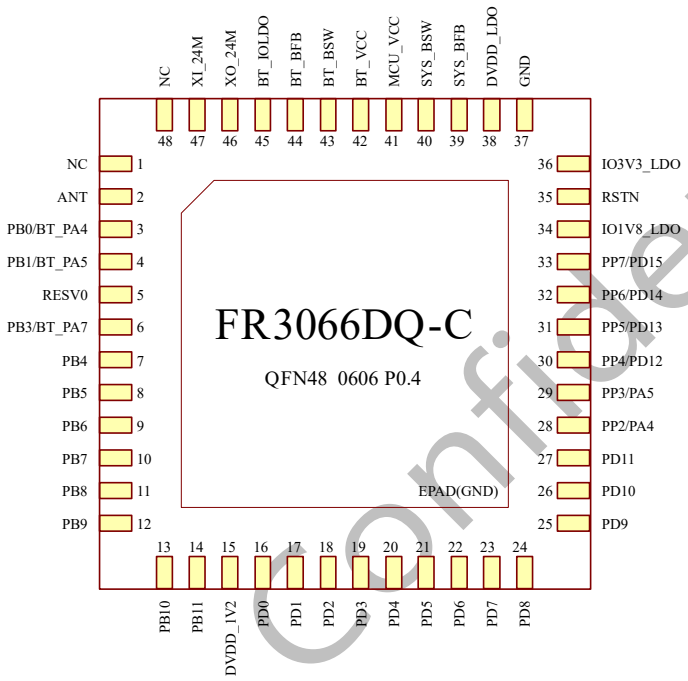
型号	工作温度	工作电压范围	闪存容量	封装	包装方式、数量	湿敏等级
FR3066DQ-C	-40°C ~ +105°C	2.9 ~ 3.6V	1MB	QFN48 6X6	编带（5000 颗 / 盘）	MSL3
FR3091E-C	-40°C ~ +85°C	2.9 ~ 3.6V	2MB	QFN68 7X7	编带（3000 颗 / 盘）	MSL3
FR5092E-C	-40°C ~ +85°C	2.9 ~ 3.6V	2MB	QFN68 7X7	编带（3000 颗 / 盘）	MSL3
FR3068E-C	-40°C ~ +85°C	2.9 ~ 3.6V	2MB	QFN80 9X9	编带（3000 颗 / 盘）	MSL3

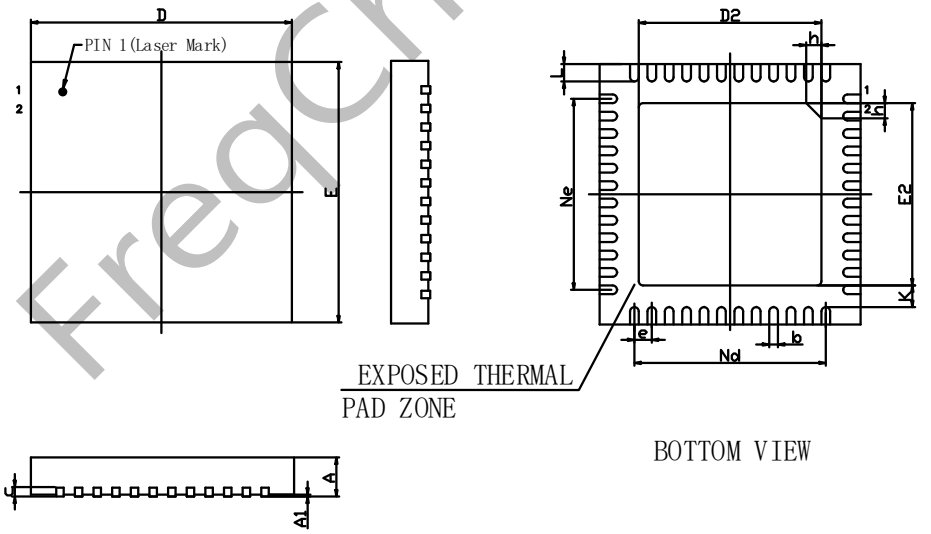
FreqChip Confidential

2. 封装定义

2.1 FR3066DQ-C 封装

- FR3066DQ-C 采用 QFN48 6X6 封装





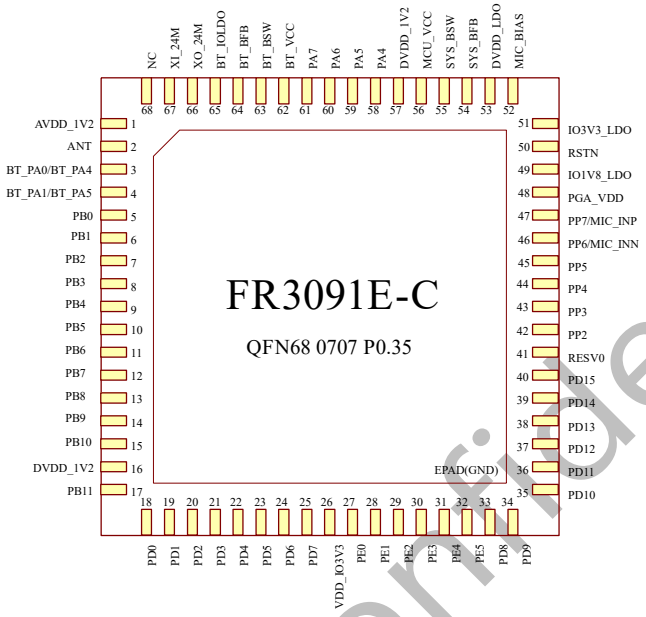
SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.85	0.90	0.95
A1	0	0.02	0.05
b	0.15	0.20	0.25
c	0.18	0.20	0.23
D	5.90	6.00	6.10
D2	4.10	4.20	4.30
e	0.40BSC		
Ne	4.40BSC		
Nd	4.40BSC		
E	5.90	6.00	6.10
E2	4.10	4.20	4.30
L	0.35	0.40	0.45
K	0.20	0.50	0.55
h	0.30	0.35	0.40
L/F载体尺寸 (MIL)	177*177		

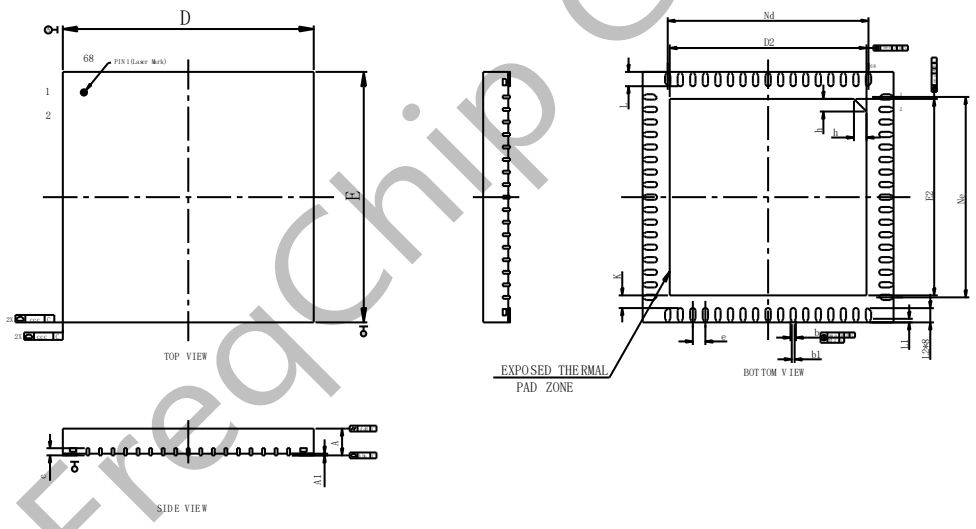
注：FR3066D-C/FR3066DQ-C/FR3066EQ-D 引脚兼容

FreqChip Confidential

2.2 FR3091E-C 封装

- FR3091E-C 采用 QFN68 7X7 封装





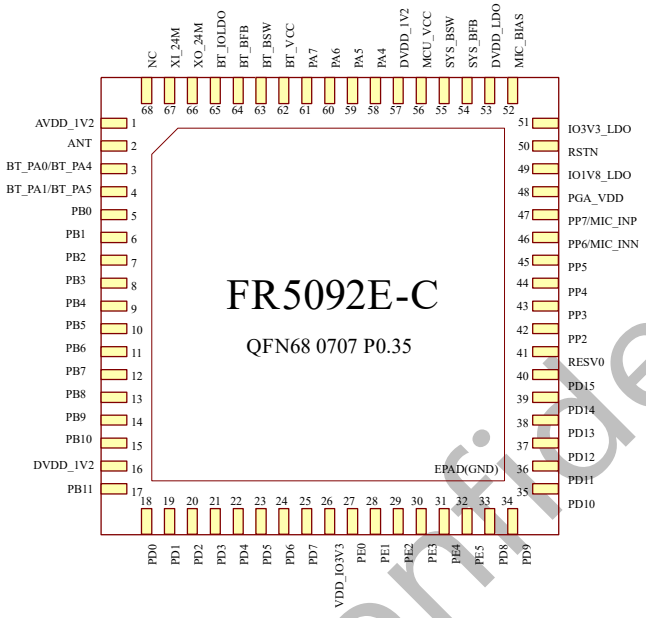
SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.80	0.85	0.90
A1	—	0.02	0.05
b	0.10	0.15	0.20
b1	0.08REF		
c	0.18	0.20	0.25
D	6.90	7.00	7.10
D2	5.39	5.49	5.59
e	0.35BSC		
Nd	5.60BSC		
E	6.90	7.00	7.10
E2	5.39	5.49	5.59
Ne	5.60BSC		
L	0.35	0.40	0.45
L1	0.10REF		
L2	0.30	0.40	0.50
K	0.20	—	—
h	0.30	0.35	0.40
aaa	0.07		
bbb	0.08		
ccc	0.10		
ddd	0.10		
eee	0.10		
fff	0.05		

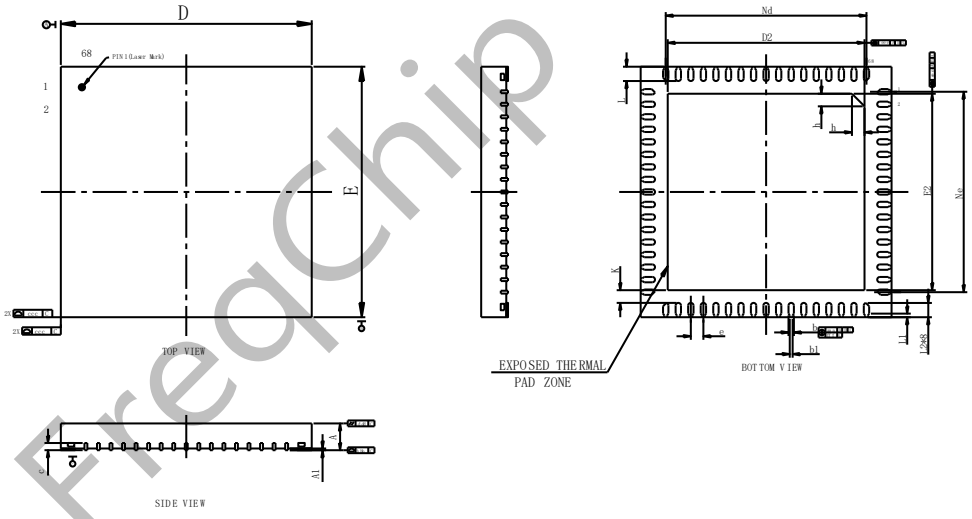
** 特殊设计：无

FreqChip Confidential

2.3 FR5092E-C 封装

- FR5092E-C 采用 QFN68 7X7 封装





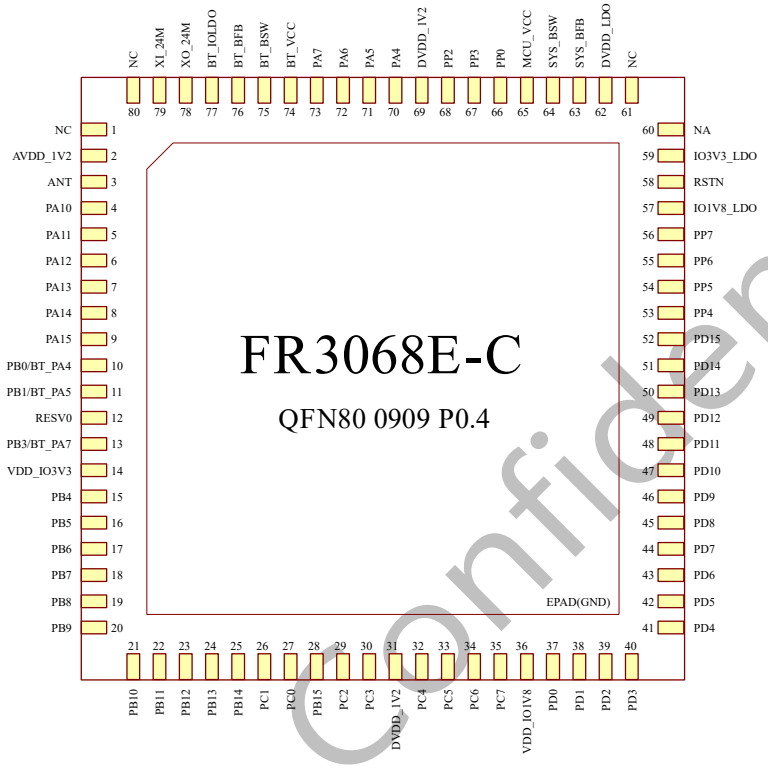
SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.80	0.85	0.90
A1	—	0.02	0.05
b	0.10	0.15	0.20
b1	0.08REF		
c	0.18	0.20	0.25
D	6.90	7.00	7.10
D2	5.39	5.49	5.59
e	0.35BSC		
Nd	5.60BSC		
E	6.90	7.00	7.10
E2	5.39	5.49	5.59
Ne	5.60BSC		
L	0.35	0.40	0.45
L1	0.10REF		
L2	0.30	0.40	0.50
K	0.20	--	--
h	0.30	0.35	0.40
aaa	0.07		
bbb	0.08		
ccc	0.10		
ddd	0.10		
eee	0.10		
fff	0.05		

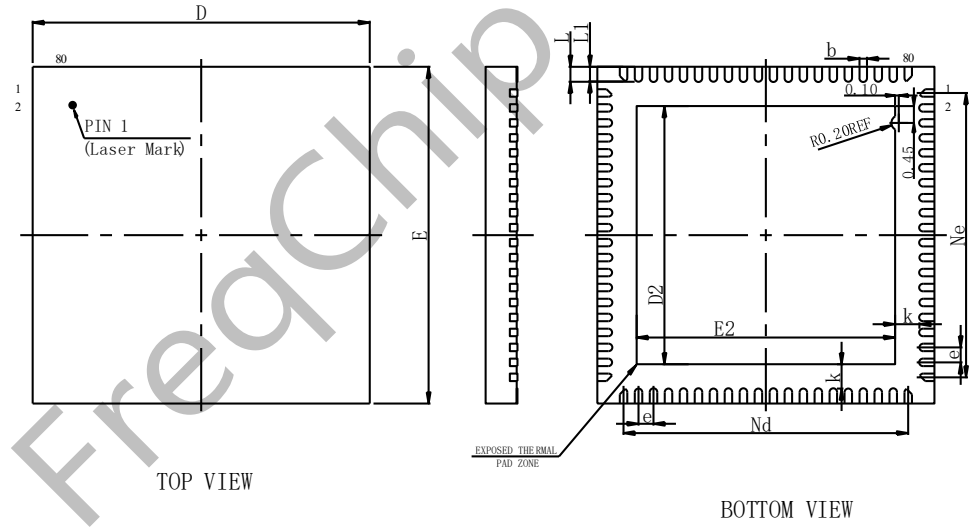
** 特殊设计：无

FreqChip Confidential

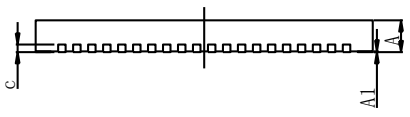
2.4 FR3068E-C 封装

- FR3068E-C 采用 QFN80 9X9 封装





SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.80	0.85	0.90
	0.85	0.90	0.95 △
	0.70	0.75	0.80 △
A1	0	0.02	0.05
b	0.15	0.20	0.25
c	0.203REF		
D	8.90	9.00	9.10
D2	6.80	6.90	7.00
e	0.40BSC		
Nd	7.60BSC		
Ne	7.60BSC		
E	8.90	9.00	9.10
E2	6.80	6.90	7.00
L	0.35	0.40	0.45
L1	0.39REF		
k	0.65REF		



**特殊设计 ：无；

注：FR 3068E-C/FR3068EP-C/FR3068E-D 引脚兼容

FreqChip Confidential

2.5 FR30xx-C 引脚定义

符号	说明
DIO	数字双向传输
AI	模拟输入
AO	模拟输出
AIO	模拟双向传输
PWR	电源
GND	地

引脚号				引脚名	类型	引脚功能
FR3066DQ-C[1]	FR3091E-C	FR5092E-C	FR3068E-C			
	1	1	2	AVDD_1V2	AI	模拟电源 1.2V 输入
2	2	2	3	ANT	AIO	外接天线接口
	58	58	70	PA4	DIO	UART1_RXD / SPIS0_SCLK / I2C1_SCK / PWM0_4 / MCAN0_RX / USB_DP
	59	59	71	PA5	DIO	UART1_TXD / SPIS0_CSN / I2C1_SDA / PWM0_5 / MCAN0_TX / SPIMX8_0_DQS / USB_DM
	60	60	72	PA6	DIO	UART1_CTS / SPIS0_MOSI / I2C0_SCK / PWM0_6 / MCAN1_RX / SPIMX8_0_SCLK / USB_DP / SPDIF_OUT
	61	61	73	PA7	DIO	UART1_RTS / SPIS0_MISO / I2C0_SDA / PWM0_7 / MCAN1_TX /

FreqChip Confidential

						SPIMX8_0_CSN / USB_DM / SPDIF_IN
			4	PA10	DIO	UART2_CTS / SPIMX2_0_MOSI / I2C1_SCK / PWM0_10 / MCAN3_RX / SPIMX8_0_IO2 / LCD_RDX / PDM0_SCK
			5	PA11	DIO	UART2_RTS / SPIMX2_0_MISO / I2C1_SDA / PWM0_11 / MCAN3_TX / SPIMX8_0_IO3 / LCD_DCX / PDM0_SDA
			6	PA12	DIO	UART1_RXD / SPIS0_SCLK / I2C0_SCK / PWM0_12 / MCAN0_RX / SPIMX8_0_IO4 / LCD_DAT0
			7	PA13	DIO	UART1_TXD / SPIS0_CSN / I2C0_SDA / PWM0_13 / MCAN0_TX / SPIMX8_0_IO5 / LCD_DAT1 / DVP_PCLK
			8	PA14	DIO	UART1_CTS / SPIS0_MOSI / I2C0_SCK / PWM0_14 / MCAN1_RX / SPIMX8_0_IO6 / LCD_DAT2 / PDM0_SCK / SPDIF_OUT / DVP_VSYNC
			9	PA15	DIO	UART1_RTS / SPIS0_MISO / I2C0_SDA / PWM0_15 / MCAN1_TX / SPIMX8_0_IO7 / LCD_DAT3 / PDM0_SDA / SPDIF_IN / DVP_HSYNC
	3	3		BT_PA0 / BT_PA4	DIO	BT_TX_IND
	4	4		BT_PA1 / BT_PA5	DIO	BT_RX_IND
3			10	PB0 / BT_PA4	DIO	UART2_RXD / SPIMX2_0_SCLK / I2C1_SCK / PWM0_0 / MCAN2_RX [2] / SPIMX8_0_SCLK / LCD_DAT4 / I2S0_SCLK BT_TX_IND / DVP_D0
4			11	PB1 / BT_PA5	DIO	UART2_TXD / SPIMX2_0_CSN / I2C1_SDA / PWM0_1 / MCAN2_TX [2] / SPIMX8_0_CSN / LCD_DAT5 / I2S0_LRCK BT_RX_IND / DVP_D1
5	41	41	12	RESV0	-	

FreqChip Confidential

6			13	PB3 / BT_PA7	DIO	UART2_RTS / SPIMX2_0_MISO / I2C0_SDA / PWM0_3 / SPIMX8_0_IO1 / LCD_DAT7 / PDM0_SDA / I2S0_DIN / SPDIF_IN / DVP_D3
	5	5		PB0	DIO	UART2_RXD / SPIMX2_0_SCLK / I2C1_SCK / PWM0_0 / MCAN2_RX / SPIMX8_0_SCLK / LCD_DAT4 / I2S0_SCLK
	6	6		PB1	DIO	UART2_TXD / SPIMX2_0_CSN / I2C1_SDA / PWM0_1 / MCAN2_TX / SPIMX8_0_CSN / LCD_DAT5 / I2S0_LRCK
	7	7		PB2	DIO	UART2_CTS / SPIMX2_0_MOSI / I2C0_SCK / PWM0_2 / MCAN3_RX / SPIMX8_0_IO0 / LCD_DAT6 / PDM0_SCK / I2S0_DOUT / SPDIF_OUT
	8	8		PB3	DIO	UART2_RTS / SPIMX2_0_MISO / I2C0_SDA / PWM0_3 / MCAN3_TX / SPIMX8_0_IO1 / LCD_DAT7 / PDM0_SDA / I2S0_DIN / SPDIF_IN
7	9	9	15	PB4	DIO	UART3_RXD / SPIS0_SCLK / I2C0_SCK / PWM0_4 / MCAN0_RX / SPIMX8_0_IO2 / LCD_DAT8 / DVP_D4
8	10	10	16	PB5	DIO	UART3_TXD / SPIS0_CSN / I2C0_SDA / PWM0_5 / MCAN0_TX / SPIMX8_0_IO3 / LCD_DAT9 / DVP_D5
9	11	11	17	PB6	DIO	UART3_CTS / SPIS0_MOSI / I2C1_SCK / PWM0_6 / MCAN1_RX / SDIOH0_CLK / SPIMX8_0_SCLK / LCD_DAT10 / PDM1_SCK / DVP_D6
10	12	12	18	PB7	DIO	UART3_RTS / SPIS0_MISO / I2C1_SDA / PWM0_7 / MCAN1_TX / SDIOH0_CMD / SPIMX8_0_CSN / LCD_DAT11 / PDM1_SDA / DVP_D7
11	13	13	19	PB8	DIO	UART2_RXD / SPIMX2_0_SCLK / I2C0_SCK / PWM0_8 / MCAN0_RX / SDIOH0_DAT0 / SPIMX8_0_IO4 / LCD_DAT12 / I2S1_SCLK / DVP_D8
12	14	14	20	PB9	DIO	UART2_TXD / SPIMX2_0_CSN / I2C0_SDA / PWM0_9 / MCAN0_TX / SDIOH0_DAT1 / SPIMX8_0_IO5 / LCD_DAT13 / I2S1_LRCK / DVP_D9
13	15	15	21	PB10	DIO	UART2_CTS / SPIMX2_0_MOSI / I2C0_SCK / PWM0_10 / MCAN1_RX /

FreqChip Confidential

						SDIOH0_DAT2 / SPIMX8_0_IO6 / LCD_DAT14 / MCU_SWCLK / PDM1_SCK / I2S1_DOUT / SPDIF_OUT / DVP_D10
14	17	17	22	PB11	DIO	UART2_RTS / SPIMX2_0_MISO / I2C0_SDA / PWM0_11 / MCAN1_TX / SDIOH0_DAT3 / SPIMX8_0_IO7 / LCD_DAT15 / MCU_SWDIO / PDM1_SDA / I2S1_DIN / SPDIF_IN / DVP_D11
			23	PB12	DIO	UART3_RXD / SPIS0_SCLK / I2C1_SCK / PWM0_12 / SPIMX8_0_IO4 / SDIOH0_DAT4 / LCD_CSX / OSPI_SCLK<XIP> / QSPI1_SCLK<XIP>
			24	PB13	DIO	UART3_TXD / SPIS0_CSN / I2C1_SDA / PWM0_13 / SPIMX8_0_IO5 / SDIOH0_DAT5 / LCD_WRX / OSPI_CSN<XIP> / QSPI1_IO0<XIP>
			25	PB14	DIO	UART3_CTS / SPIS0_MOSI / I2C0_SCK / PWM0_14 / SPIMX8_0_IO6 / SDIOH0_DAT6 / LCD_RDX / OSPI_DQS<XIP> / PDM1_SCK / QSPI1_IO3<XIP>
			28	PB15	DIO	UART3_RTS / SPIS0_MISO / I2C0_SDA / PWM0_15 / SPIMX8_0_IO7 / SDIOH0_DAT7 / LCD_DCX / OSPI_IO7<XIP> / PDM1_SDA / QSPI1_IO2<XIP>
			27	PC0	DIO	UART2_RXD / SPIMX2_0_SCLK / I2C1_SCK / PWM1_0 / SPIMX8_0_IO0 / SDIOH0_DAT0 / LCD_DAT0 / OSPI_IO6<XIP> / I2S1_SCLK / QSPI1_IO1<XIP>
			26	PC1	DIO	UART2_TXD / SPIMX2_0_CSN / I2C1_SDA / PWM1_1 / SPIMX8_0_IO1 / SDIOH0_DAT1 / LCD_DAT1 / OSPI_IO5<XIP> / I2S1_LRCK / QSPI1_CSN<XIP>
			29	PC2	DIO	UART2_CTS / SPIMX2_0_MOSI / I2C2_SCK / PWM1_2 / SPIMX8_0_IO2 / SDIOH0_DAT2 / LCD_DAT2 / OSPI_IO1<XIP> / PDM1_SCK / I2S1_DOUT / SPDIF_OUT
			30	PC3	DIO	UART2_RTS / SPIMX2_0_MISO / I2C2_SDA / PWM1_3 / SPIMX8_0_IO3 / SDIOH0_DAT3 / LCD_DAT3 / OSPI_IO3<XIP> / PDM1_SDA / I2S1_DIN / SPDIF_IN

FreqChip Confidential

	26	26	14	VDD_IO3V3	PWR	3.3V IO 供电
			36	VDD_IO1V8	PWR	1.8V IO 供电
			32	PC4	DIO	UART3_RXD / SPIS0_SCLK / I2C2_SCK / PWM1_4 / SPIMX8_0_SCLK / SDIOH0_CLK / LCD_DAT4 / OSPI_IO4<XIP> / DAC_PWMLP/PWM_LCK
			33	PC5	DIO	UART3_TXD / SPIS0_CSN / I2C2_SDA / PWM1_5 / SPIMX8_0_CSN / SDIOH0_CMD / SPIMX8_1_DQS / LCD_DAT5 / OSPI_IO2<XIP> / DAC_PWMLN/PWM_LCK
			34	PC6	DIO	UART3_CTS / SPIS0_MOSI / I2C1_SCK / PWM1_6 / SPIMX8_0_DQS / SDIOH0_DAT4 / SPIMX8_1_IO4 / LCD_DAT6 / OSPI_IO0<XIP> / PDM1_SCK / DAC_PWMRP/PWM_RCK
			35	PC7	DIO	UART3_RTS / SPIS0_MISO / I2C1_SDA / PWM1_7 / SDIOH0_DAT5 / SPIMX8_1_IO5 / LCD_DAT7 / OSPI_SCLK#<XIP> / PDM1_SDA / DAC_PWMRN/PWM_RCK
16	18	18	37	PD0	DIO	UART4_RXD / SPIMX2_1_SCLK / I2C2_SCK / PWM1_0 / MCAN0_RX / SDIOH0_CLK / SPIMX8_1_SCLK / LCD_DAT0 / I2S1_SCLK / DVP_D0
17	19	19	38	PD1	DIO	UART4_TXD / SPIMX2_1_CSN / I2C2_SDA / PWM1_1 / MCAN0_TX / SDIOH0_CMD / SPIMX8_1_CSN / LCD_DAT1 / I2S1_LRCK / DVP_D1
18	20	20	39	PD2	DIO	UART4_CTS / SPIMX2_1_MOSI / I2C1_SCK / PWM1_2 / MCAN1_RX / SDIOH0_DAT4 / SPIMX8_1_IO0 / LCD_DAT2 / PDM1_SCK / I2S1_DOUT / SPDIF_OUT / DVP_D2
19	21	21	40	PD3	DIO	UART4_RTS / SPIMX2_1_MISO / I2C1_SDA / PWM1_3 / MCAN1_TX / SDIOH0_DAT5 / SPIMX8_1_IO1 / LCD_DAT3 / PDM1_SDA / I2S1_DIN /

FreqChip Confidential

						SPDIF_IN / DVP_D3
20	22	22	41	PD4	DIO	UART5_RXD / I2C2_SCK / PWM1_4 / SPIS1_SCLK / SDIOH0_DAT6 / SPIMX8_1_IO2 / LCD_DAT4 / DAC_PWMLP/PWM_LCK / DVP_D4
21	23	23	42	PD5	DIO	UART5_TXD / I2C2_SDA / PWM1_5 / SPIS1_CSN / SDIOH0_DAT7 / SPIMX8_1_IO3 / LCD_DAT5 / DAC_PWMLN/PWM_LCK / DVP_D5
22	24	24	43	PD6	DIO	UART5_CTS / I2C2_SCK / PWM1_6 / SPIS1_IO0 / SDIOH0_DAT0 / SPIMX8_1_IO4 / LCD_DAT6 / PDM1_SCK / DAC_PWMRP/PWM_RCK / DVP_D6
23	25	25	44	PD7	DIO	UART5_RTS / I2C2_SDA / PWM1_7 / SPIS1_IO1 / SDIOH0_DAT1 / SPIMX8_1_IO5 / LCD_DAT7 / PDM1_SDA / DAC_PWMRN/PWM_RCK / DVP_D7
24	33	33	45	PD8	DIO	UART4_RXD / SPIMX2_1_SCLK / I2C1_SCK / PWM1_8 / SPIS1_IO2 / SDIOH0_DAT2 / SPIMX8_1_IO6 / LCD_CSX / I2S0_SCLK / DAC_PWMCLK / DVP_D8
25	34	34	46	PD9	DIO	UART4_TXD / SPIMX2_1_CSN / I2C1_SDA / PWM1_9 / SPIS1_IO3 / SDIOH0_DAT3 / SPIMX8_1_IO7 / LCD_WRX / I2S0_LRCK / DVP_D9
26	35	35	47	PD10	DIO	UART4_CTS / SPIMX2_1_MOSI / I2C2_SCK / PWM1_10 / SPIS1_SCLK / SDIOH0_CLK / SPIMX8_1_DQS / LCD_RDX / PDM1_SCK / I2S0_DOUT / SPDIF_OUT / DVP_D10
27	36	36	48	PD11	DIO	UART4_RTS / SPIMX2_1_MISO / I2C2_SDA / PWM1_11 / SPIS1_CSN / SDIOH0_CMD / LCD_DCX / PDM1_SDA / I2S0_DIN / SPDIF_IN / DAC_PWMLP/PWM_LCK / DVP_D11
	37	37	49	PD12	DIO	UART5_RXD / I2C2_SCK / PWM1_12 / MCAN0_RX / SDIOH0_DAT4 /

FreqChip Confidential

						LCD_DAT8 / USB_DP / I2S1_SCLK / DAC_PWMLN/PWM_LCK
	38	38	50	PD13	DIO	UART5_TXD / I2C2_SDA / PWM1_13 / MCAN0_TX / SDIOH0_DAT5 / LCD_DAT9 / USB_DM / I2S1_LRCK / DAC_PWMRP/PWM_RCK
	39	39	51	PD14	DIO	UART5_CTS / I2C1_SCK / PWM1_14 / MCAN1_RX / SDIOH0_DAT6 / LCD_DAT10 / USB_DP / PDM1_SCK / I2S1_DOUT / SPDIF_OUT / DAC_PWMRN/PWM_RCK
	40	40	52	PD15	DIO	UART5_RTS / I2C1_SDA / PWM1_15 / MCAN1_TX / SDIOH0_DAT7 / LCD_DAT11 / USB_DM / PDM1_SDA / I2S1_DIN / SPDIF_IN / DAC_PWMCLK
	27	27		PE0	DIO	UART3_RXD / SPIMX2_1_SCLK / I2C2_SCK / PWM1_0 / MCAN2_RX / SDIOH0_CLK / SPIMX8_1_IO0 / LCD_DAT12 / PDM0_SCK / I2S0_SCLK / SPDIF_OUT / DAC_PWMLP/PWM_LCK / DVP_PCLK
	28	28		PE1	DIO	UART3_TXD / SPIMX2_1_CSN / I2C2_SDA / PWM1_1 / MCAN2_TX / SDIOH0_CMD / SPIMX8_1_IO1 / LCD_DAT13 / PDM0_SDA / I2S0_LRCK / SPDIF_IN / DAC_PWMLN/PWM_LCK / DVP_VSYNC
	29	29		PE2	DIO	UART3_CTS / SPIMX2_1_MOSI / I2C1_SCK / PWM1_2 / MCAN3_RX / SDIOH0_DAT0 / SPIMX8_1_IO2 / LCD_DAT14 / I2S0_DOUT / DAC_PWMRP/PWM_RCK / DVP_HSYNC
	30	30		PE3	DIO	UART3_RTS / SPIMX2_1_MISO / I2C1_SDA / PWM1_3 / MCAN3_TX / SDIOH0_DAT1 / SPIMX8_1_IO3 / LCD_DAT15 / I2S0_DIN / DAC_PWMRN/PWM_RCK
	31	31		PE4	DIO	UART4_RXD / I2C2_SCK / PWM1_4 / SPIS1_SCLK / SDIOH0_DAT2 / SPIMX8_1_SCLK / LCD_DAT16 / PDM0_SCK / I2S1_SCLK / DAC_PWMCLK
	32	32		PE5	DIO	UART4_TXD / I2C2_SDA / PWM1_5 / SPIS1_CSN / SDIOH0_DAT3 /

FreqChip Confidential

						SPIMX8_1_CSN / LCD_DAT17 / PDM0_SDA / I2S1_LRCK
28				PP2 / PA4	DIO	PMU_IO2/ADC UART2_RXD / SPIMX2_0_SCLK / I2C1_SCK / PWM0_0 / MCAN2_RX / SPIMX8_0_SCLK / LCD_DAT4 / I2S0_SCLK BT_TX_IND
29				PP3 / PA5	DIO	PMU_IO3/ADC UART2_TXD / SPIMX2_0_CSN / I2C1_SDA / PWM0_1 / MCAN2_TX / SPIMX8_0_CSN / LCD_DAT5 / I2S0_LRCK
30				PP4 / PD12	DIO	PMU_IO4/ADC UART5_RXD / I2C2_SCK / PWM1_12 / SDIOH0_DAT4 / LCD_DAT8 / USB_DP / I2S1_SCLK / DAC_PWMLN/PWM_LCK
31				PP5 / PD13	DIO	PMU_IO5 UART5_TXD / I2C2_SDA / PWM1_13 / SDIOH0_DAT5 / LCD_DAT9 / USB_DM / I2S1_LRCK / DAC_PWMRP/PWM_RCK
32				PP6 / PD14	DIO	PMU_IO6 UART5_CTS / I2C1_SCK / PWM1_14 / MCAN1_RX / SDIOH0_DAT6 / LCD_DAT10 / USB_DP / PDM1_SCK / I2S1_DOUT / SPDIF_OUT / DAC_PWMRN/PWM_RCK
33				PP7 / PD15	DIO	PMU_IO7 UART5_RTS / I2C1_SDA / PWM1_15 / MCAN1_TX / SDIOH0_DAT7 / LCD_DAT11 / USB_DM / PDM1_SDA / I2S1_DIN / SPDIF_IN / DAC_PWMCLK
			66	PP0	DIO	PMU_IO0/ADC/ADKEY0

FreqChip Confidential

	42	42	68	PP2	DIO	PMU_IO2/ADC
	43	43	67	PP3	DIO	PMU_IO3/ADC
	44	44	53	PP4	DIO	PMU_IO4/ADC
	45	45	54	PP5	DIO	PMU_IO5
			55	PP6	DIO	PMU_IO6
			56	PP7	DIO	PMU_IO7
	46	46		PP6 / MIC_INN	DIO	PMU_IO6/PGA_INN
	47	47		PP7 / MIC_INP	DIO	PMU_IO7/PGA_INP
	48	48		PGA_VDD	PWR	麦克风可编程增益放大器电源
34	49	49	57	IO1V8_LDO	PWR	1.8V IO LDO
35	50	50	58	RSTN	AI	芯片复位引脚，低有效
36	51	51	59	IO3V3_LDO	PWR	3.3V IO LDO
	52	52		MIC_BAIS	PWR	麦克风偏置电压输出
38	53	53	62	DVDD_LDO	PWR	数字 1.2V LDO 输出
39	54	54	63	SYS_BFB	PWR	系统 BUCK 电路反馈输入
40	55	55	64	SYS_BSW	PWR	系统 BUCK 电路输出
41	56	56	65	MCU_VCC	PWR	MCU 供电

Freqchip Confidential

15	16/57	16/57	31/69	DVDD_1V2	PWR	数字 1.2V 电源
42	62	62	74	BT_VCC	PWR	蓝牙供电
43	63	63	75	BT_BSW	AO	Buck 电路输出
44	64	64	76	BT_BFB	AI	Buck 电路反馈输入端
45	65	65	77	BT_IOLDO	AO	用于 GPIO 供电
46	66	66	78	XO_24M	AO	24MHz 晶振输出端
47	67	67	79	XI_24M	AI	24MHz 晶振输入端
37/49	69	69	81	GND	GND	
1/ 48	68	68	1 / 61 / 80	NC	-	NC
			60	N/A	-	NA

FreqChip Confidential

注释：

[1] FR3066x-C 系列不支持 DVP 功能；

[2] MCAN2 只有 FR3066EQ-D 支持（该型号支持 3 路 CAN）

2.6 FR3066D-C/FR3066DQ-C/FR3066EQ-C 系列 GPIO 功能复用表

名称	GPIO 功能													
PB0/BT_PA4	GPIOB_0	UART2_RXD	SPIMX2_0_SCLK	I2C1_SCK	PWM0_0	MCAN2_RX[2]		SPIMX8_0_SCLK	LCD_DAT4			I2S0_SCLK		
	BT_GPIOA_4	BT_TX_IND												
PB1/BT_PA5	GPIOB_1	UART2_TXD	SPIMX2_0_CSN	I2C1_SDA	PWM0_1	MCAN2_TX[2]		SPIMX8_0_CSN	LCD_DAT5			I2S0_LRCK		
	BT_GPIOA_5	BT_RX_IND												
PB3/BT_PA7	GPIOB_3	UART2_RTS	SPIMX2_0_MISO	I2C0_SDA	PWM0_3			SPIMX8_0_IO1	LCD_DAT7		PDM0_SDA	I2S0_DIN	SPDIF_IN	
	BT_GPIOA_7													
PB4	GPIOB_4	UART3_RXD	SPIS0_SCLK	I2C0_SCK	PWM0_4	MCAN0_RX		SPIMX8_0_IO2	LCD_DAT8					
PB5	GPIOB_5	UART3_TXD	SPIS0_CSN	I2C0_SDA	PWM0_5	MCAN0_TX		SPIMX8_0_IO3	LCD_DAT9					
PB6	GPIOB_6	UART3_CTS	SPIS0_MOSI	I2C1_SCK	PWM0_6	MCAN1_RX	SDIOH0_CLK	SPIMX8_0_SCLK	LCD_DAT10		PDM1_SCK			
PB7	GPIOB_7	UART3_RTS	SPIS0_MISO	I2C1_SDA	PWM0_7	MCAN1_TX	SDIOH0_CMD	SPIMX8_0_CSN	LCD_DAT11		PDM1_SDA			
PB8	GPIOB_8	UART2_RXD	SPIMX2_0_SCLK	I2C0_SCK	PWM0_8	MCAN0_RX	SDIOH0_DAT0	SPIMX8_0_IO4	LCD_DAT12			I2S1_SCLK		
PB9	GPIOB_9	UART2_TXD	SPIMX2_0_CSN	I2C0_SDA	PWM0_9	MCAN0_TX	SDIOH0_DAT1	SPIMX8_0_IO5	LCD_DAT13			I2S1_LRCK		
PB10	GPIOB_10	UART2_CTS	SPIMX2_0_MOSI	I2C0_SCK	PWM0_10	MCAN1_RX	SDIOH0_DAT2	SPIMX8_0_IO6	LCD_DAT14	MCU_SWCLK	PDM1_SCK	I2S1_DOUT	SPDIF_OUT	
PB11	GPIOB_11	UART2_RTS	SPIMX2_0_MISO	I2C0_SDA	PWM0_11	MCAN1_TX	SDIOH0_DAT3	SPIMX8_0_IO7	LCD_DAT15	MCU_SWDIO	PDM1_SDA	I2S1_DIN	SPDIF_IN	
PD0	GPIOD_0	UART4_RXD	SPIMX2_1_SCLK	I2C2_SCK	PWM1_0	MCAN0_RX	SDIOH0_CLK	SPIMX8_1_SCLK	LCD_DAT0			I2S1_SCLK		
PD1	GPIOD_1	UART4_TXD	SPIMX2_1_CSN	I2C2_SDA	PWM1_1	MCAN0_TX	SDIOH0_CMD	SPIMX8_1_CSN	LCD_DAT1			I2S1_LRCK		
PD2	GPIOD_2	UART4_CTS	SPIMX2_1_MOSI	I2C1_SCK	PWM1_2	MCAN1_RX	SDIOH0_DAT4	SPIMX8_1_IO0	LCD_DAT2		PDM1_SCK	I2S1_DOUT	SPDIF_OUT	
PD3	GPIOD_3	UART4_RTS	SPIMX2_1_MISO	I2C1_SDA	PWM1_3	MCAN1_TX	SDIOH0_DAT5	SPIMX8_1_IO1	LCD_DAT3		PDM1_SDA	I2S1_DIN	SPDIF_IN	

Confidential

名称	GPIO 功能													
PD4	GPIOD_4	UART5_RXD		I2C2_SCK	PWM1_4	SPIS1_SCLK	SDIOH0_DAT6	SPIMX8_1_IO2	LCD_DAT4					DAC_PWMLP/PWM_LCK
PD5	GPIOD_5	UART5_TXD		I2C2_SDA	PWM1_5	SPIS1_CSN	SDIOH0_DAT7	SPIMX8_1_IO3	LCD_DAT5					DAC_PWMLN/PWM_LCK
PD6	GPIOD_6	UART5_CTS		I2C2_SCK	PWM1_6	SPIS1_IO0	SDIOH0_DAT0	SPIMX8_1_IO4	LCD_DAT6		PDM1_SCK			DAC_PWMRP/PWM_RCK
PD7	GPIOD_7	UART5_RTS		I2C2_SDA	PWM1_7	SPIS1_IO1	SDIOH0_DAT1	SPIMX8_1_IO5	LCD_DAT7		PDM1_SDA			DAC_PWMRN/PWM_RCK
PD8	GPIOD_8	UART4_RXD	SPIMX2_1_SCLK	I2C1_SCK	PWM1_8	SPIS1_IO2	SDIOH0_DAT2	SPIMX8_1_IO6	LCD_CSX			I2S0_SCLK		DAC_PWMCLK
PD9	GPIOD_9	UART4_TXD	SPIMX2_1_CSN	I2C1_SDA	PWM1_9	SPIS1_IO3	SDIOH0_DAT3	SPIMX8_1_IO7	LCD_WRX			I2S0_LRCK		
PD10	GPIOD_10	UART4_CTS	SPIMX2_1_MOSI	I2C2_SCK	PWM1_10	SPIS1_SCLK	SDIOH0_CLK	SPIMX8_1_DQS	LCD_RDX		PDM1_SCK	I2S0_DOUT	SPDIF_OUT	
PD11	GPIOD_11/ADC	UART4_RTS	SPIMX2_1_MISO	I2C2_SDA	PWM1_11	SPIS1_CSN	SDIOH0_CMD		LCD_DCX		PDM1_SDA	I2S0_DIN	SPDIF_IN	DAC_PWMLP/PWM_LCK
PP2/PA4	PMU_IO2/ADC													
	GPIOA_4	UART1_RXD	SPIS0_SCLK	I2C1_SCK	PWM0_4					USB_DP				
PP3/PA5	PMU_IO3/ADC													
	GPIOA_5	UART1_TXD	SPIS0_CSN	I2C1_SDA	PWM0_5			SPIMX8_0_DQS		USB_DM				
PP4/PD12	PMU_IO4													
	GPIOD_12/ADC	UART5_RXD		I2C2_SCK	PWM1_12		SDIOH0_DAT4		LCD_DAT8	USB_DP		I2S1_SCLK		DAC_PWMLN/PWM_LCK
PP5/PD13	PMU_IO5/ADC													
	GPIOD_13/ADC	UART5_TXD		I2C2_SDA	PWM1_13		SDIOH0_DAT5		LCD_DAT9	USB_DM		I2S1_LRCK		DAC_PWMRP/PWM_RCK
PP6/PD14	PMU_IO6/PGA_INN													
	GPIOD_14/ADC	UART5_CTS		I2C1_SCK	PWM1_14	MCAN1_RX	SDIOH0_DAT6		LCD_DAT10	USB_DP	PDM1_SCK	I2S1_DOUT	SPDIF_OUT	DAC_PWMRN/PWM_RCK
PP7/PD15	PMU_IO7/PGA_INP													
	GPIOD_15/ADC	UART5_RTS		I2C1_SDA	PWM1_15	MCAN1_TX	SDIOH0_DAT7		LCD_DAT11	USB_DM	PDM1_SDA	I2S1_DIN	SPDIF_IN	DAC_PWMCLK

FreqChip Confidential

2.7 FR3091E-C/ FR5092E-C 系列 GPIO 功能复用表

名称	GPIO 功能														
BT_PA0/BT_PA4	BT_GPIOA_0 & BT_GPIO_4	BT_TX_IND													
BT_PA1/BT_PA5	BT_GPIOA_1 & BT_GPIO_5	BT_RX_IND													
PB0	GPIOB_0	UART2_RXD	SPIMX2_0_SCLK	I2C1_SCK	PWM0_0	MCAN2_RX		SPIMX8_0_SCLK	LCD_DAT4			I2S0_SCLK			DVP_D0
PB1	GPIOB_1	UART2_TXD	SPIMX2_0_CSN	I2C1_SDA	PWM0_1	MCAN2_TX		SPIMX8_0_CSN	LCD_DAT5			I2S0_LRCK			DVP_D1
PB2	GPIOB_2	UART2_CTS	SPIMX2_0_MOSI	I2C0_SCK	PWM0_2	MCAN3_RX		SPIMX8_0_IO0	LCD_DAT6		PDM0_SCK	I2S0_DOUT	SPDIF_OUT		DVP_D2
PB3	GPIOB_3	UART2_RTS	SPIMX2_0_MISO	I2C0_SDA	PWM0_3	MCAN3_TX		SPIMX8_0_IO1	LCD_DAT7		PDM0_SDA	I2S0_DIN	SPDIF_IN		DVP_D3
PB4	GPIOB_4	UART3_RXD	SPIS0_SCLK	I2C0_SCK	PWM0_4	MCAN0_RX		SPIMX8_0_IO2	LCD_DAT8						DVP_D4
PB5	GPIOB_5	UART3_TXD	SPIS0_CSN	I2C0_SDA	PWM0_5	MCAN0_TX		SPIMX8_0_IO3	LCD_DAT9						DVP_D5
PB6	GPIOB_6	UART3_CTS	SPIS0_MOSI	I2C1_SCK	PWM0_6	MCAN1_RX	SDIOH0_CLK	SPIMX8_0_SCLK	LCD_DAT10		PDM1_SCK				DVP_D6
PB7	GPIOB_7	UART3_RTS	SPIS0_MISO	I2C1_SDA	PWM0_7	MCAN1_TX	SDIOH0_CMD	SPIMX8_0_CSN	LCD_DAT11		PDM1_SDA				DVP_D7
PB8	GPIOB_8	UART2_RXD	SPIMX2_0_SCLK	I2C0_SCK	PWM0_8	MCAN0_RX	SDIOH0_DAT0	SPIMX8_0_IO4	LCD_DAT12			I2S1_SCLK			DVP_D8
PB9	GPIOB_9	UART2_TXD	SPIMX2_0_CSN	I2C0_SDA	PWM0_9	MCAN0_TX	SDIOH0_DAT1	SPIMX8_0_IO5	LCD_DAT13			I2S1_LRCK			DVP_D9
PB10	GPIOB_10	UART2_CTS	SPIMX2_0_MOSI	I2C0_SCK	PWM0_10	MCAN1_RX	SDIOH0_DAT2	SPIMX8_0_IO6	LCD_DAT14	MCU_SWCLK	PDM1_SCK	I2S1_DOUT	SPDIF_OUT		DVP_D10
PB11	GPIOB_11	UART2_RTS	SPIMX2_0_MISO	I2C0_SDA	PWM0_11	MCAN1_TX	SDIOH0_DAT3	SPIMX8_0_IO7	LCD_DAT15	MCU_SWDIO	PDM1_SDA	I2S1_DIN	SPDIF_IN		DVP_D11
PD0	GPIOD_0	UART4_RXD	SPIMX2_1_SCLK	I2C2_SCK	PWM1_0	MCAN0_RX	SDIOH0_CLK	SPIMX8_1_SCLK	LCD_DAT0			I2S1_SCLK			DVP_D0
PD1	GPIOD_1	UART4_TXD	SPIMX2_1_CSN	I2C2_SDA	PWM1_1	MCAN0_TX	SDIOH0_CMD	SPIMX8_1_CSN	LCD_DAT1			I2S1_LRCK			DVP_D1
PD2	GPIOD_2	UART4_CTS	SPIMX2_1_MOSI	I2C1_SCK	PWM1_2	MCAN1_RX	SDIOH0_DAT4	SPIMX8_1_IO0	LCD_DAT2		PDM1_SCK	I2S1_DOUT	SPDIF_OUT		DVP_D2

Confidential

名称	GPIO 功能														
PD3	GPIOD_3	UART4_RTS	SPIMX2_1_MISO	I2C1_SDA	PWM1_3	MCAN1_TX	SDIOH0_DAT5	SPIMX8_1_IO1	LCD_DAT3		PDM1_SDA	I2S1_DIN	SPDIF_IN		DVP_D3
PD4	GPIOD_4	UART5_RXD		I2C2_SCK	PWM1_4	SPIS1_SCLK	SDIOH0_DAT6	SPIMX8_1_IO2	LCD_DAT4					DAC_PWMLP/PWM_LCK	DVP_D4
PD5	GPIOD_5	UART5_TXD		I2C2_SDA	PWM1_5	SPIS1_CSN	SDIOH0_DAT7	SPIMX8_1_IO3	LCD_DAT5					DAC_PWMLN/PWM_LCK	DVP_D5
PD6	GPIOD_6	UART5_CTS		I2C2_SCK	PWM1_6	SPIS1_IO0	SDIOH0_DAT0	SPIMX8_1_IO4	LCD_DAT6		PDM1_SCK			DAC_PWMRP/PWM_RCK	DVP_D6
PD7	GPIOD_7	UART5_RTS		I2C2_SDA	PWM1_7	SPIS1_IO1	SDIOH0_DAT1	SPIMX8_1_IO5	LCD_DAT7		PDM1_SDA			DAC_PWMRN/PWM_RCK	DVP_D7
PE0	GPIOE_0	UART3_RXD	SPIMX2_1_SCLK	I2C2_SCK	PWM1_0	MCAN2_RX	SDIOH0_CLK	SPIMX8_1_IO0	LCD_DAT12		PDM0_SCK	I2S0_SCLK	SPDIF_OUT	DAC_PWMLP/PWM_LCK	DVP_PCLK
PE1	GPIOE_1	UART3_TXD	SPIMX2_1_CSN	I2C2_SDA	PWM1_1	MCAN2_TX	SDIOH0_CMD	SPIMX8_1_IO1	LCD_DAT13		PDM0_SDA	I2S0_LRCK	SPDIF_IN	DAC_PWMLN/PWM_LCK	DVP_VSYNC
PE2	GPIOE_2	UART3_CTS	SPIMX2_1_MOSI	I2C1_SCK	PWM1_2	MCAN3_RX	SDIOH0_DAT0	SPIMX8_1_IO2	LCD_DAT14			I2S0_DOUT		DAC_PWMRP/PWM_RCK	DVP_HSYNC
PE3	GPIOE_3	UART3_RTS	SPIMX2_1_MISO	I2C1_SDA	PWM1_3	MCAN3_TX	SDIOH0_DAT1	SPIMX8_1_IO3	LCD_DAT15			I2S0_DIN		DAC_PWMRN/PWM_RCK	
PE4	GPIOE_4	UART4_RXD		I2C2_SCK	PWM1_4	SPIS1_SCLK	SDIOH0_DAT2	SPIMX8_1_SCLK	LCD_DAT16		PDM0_SCK	I2S1_SCLK		DAC_PWMCLK	
PE5	GPIOE_5	UART4_TXD		I2C2_SDA	PWM1_5	SPIS1_CSN	SDIOH0_DAT3	SPIMX8_1_CSN	LCD_DAT17		PDM0_SDA	I2S1_LRCK			
PD8	GPIOD_8	UART4_RXD	SPIMX2_1_SCLK	I2C1_SCK	PWM1_8	SPIS1_IO2	SDIOH0_DAT2	SPIMX8_1_IO6	LCD_CSX			I2S0_SCLK		DAC_PWMCLK	DVP_D8
PD9	GPIOD_9	UART4_TXD	SPIMX2_1_CSN	I2C1_SDA	PWM1_9	SPIS1_IO3	SDIOH0_DAT3	SPIMX8_1_IO7	LCD_WRX			I2S0_LRCK			DVP_D9
PD10	GPIOD_10	UART4_CTS	SPIMX2_1_MOSI	I2C2_SCK	PWM1_10	SPIS1_SCLK	SDIOH0_CLK	SPIMX8_1_DQS	LCD_RDX		PDM1_SCK	I2S0_DOUT	SPDIF_OUT		DVP_D10
PD11	GPIOD_11/ADC	UART4_RTS	SPIMX2_1_MISO	I2C2_SDA	PWM1_11	SPIS1_CSN	SDIOH0_CMD		LCD_DCX		PDM1_SDA	I2S0_DIN	SPDIF_IN	DAC_PWMLP/PWM_LCK	DVP_D11
PD12	GPIOD_12/ADC	UART5_RXD		I2C2_SCK	PWM1_12	MCAN0_RX	SDIOH0_DAT4		LCD_DAT8	USB_DP		I2S1_SCLK		DAC_PWMLN/PWM_LCK	
PD13	GPIOD_13/ADC	UART5_TXD		I2C2_SDA	PWM1_13	MCAN0_TX	SDIOH0_DAT5		LCD_DAT9	USB_DM		I2S1_LRCK		DAC_PWMRP/PWM_RCK	
PD14	GPIOD_14/ADC	UART5_CTS		I2C1_SCK	PWM1_14	MCAN1_RX	SDIOH0_DAT6		LCD_DAT10	USB_DP	PDM1_SCK	I2S1_DOUT	SPDIF_OUT	DAC_PWMRN/PWM_RCK	
PD15	GPIOD_15/ADC	UART5_RTS		I2C1_SDA	PWM1_15	MCAN1_TX	SDIOH0_DAT7		LCD_DAT11	USB_DM	PDM1_SDA	I2S1_DIN	SPDIF_IN	DAC_PWMCLK	
PP2	PMU_IO2/ADC														

Confidential

名称	GPIO 功能														
PP3	PMU_IO3/ADC														
PP4	PMU_IO4/ADC														
PP5	PMU_IO5														
PP6	PMU_IO6/PGA_IN N														
PP7	PMU_IO7/PGA_INP														
PA4	GPIOA_4	UART1_ RXD	SPIS0_SCLK	I2C1_S CK	PWM0 _4	MCAN0 _RX				USB_DP					
PA5	GPIOA_5	UART1_ TXD	SPIS0_CSN	I2C1_S DA	PWM0 _5	MCAN0 _TX		SPIMX8_0_ DQS		USB_DM					
PA6	GPIOA_6	UART1_ CTS	SPIS0_MOSI	I2C0_S CK	PWM0 _6	MCAN1 _RX		SPIMX8_0_ SCLK		USB_DP			SPDIF_ OUT		
PA7	GPIOA_7	UART1_ RTS	SPIS0_MISO	I2C0_S DA	PWM0 _7	MCAN1 _TX		SPIMX8_0_ CSN		USB_DM			SPDIF_I N		

FreqChip Confidential

2.8 FR3068E-C/FR3068EP-C/FR3068E-D 系列 GPIO 功能复用表

名称	GPIO 功能														
PA10	GPIOA_10	UART2_CTS	SPIMX2_0_MOSI	I2C1_SCK	PWM0_10	MCAN3_RX		SPIMX8_0_IO2	LCD_RDX		PDM0_SCK				
PA11	GPIOA_11	UART2_RTS	SPIMX2_0_MISO	I2C1_SDA	PWM0_11	MCAN3_TX		SPIMX8_0_IO3	LCD_DCX		PDM0_SDA				
PA12	GPIOA_12	UART1_RXD	SPIS0_SCLK	I2C0_SCK	PWM0_12	MCAN0_RX		SPIMX8_0_IO4	LCD_DAT0						
PA13	GPIOA_13	UART1_TXD	SPIS0_CSN	I2C0_SDA	PWM0_13	MCAN0_TX		SPIMX8_0_IO5	LCD_DAT1						DVP_PCLK
PA14	GPIOA_14	UART1_CTS	SPIS0_MOSI	I2C0_SCK	PWM0_14	MCAN1_RX		SPIMX8_0_IO6	LCD_DAT2		PDM0_SCK		SPDIF_OUT		DVP_VSYNC
PA15	GPIOA_15	UART1_RTS	SPIS0_MISO	I2C0_SDA	PWM0_15	MCAN1_TX		SPIMX8_0_IO7	LCD_DAT3		PDM0_SDA		SPDIF_IN		DVP_HSYNC
PB0/BT_PA4	GPIOB_0	UART2_RXD	SPIMX2_0_SCLK	I2C1_SCK	PWM0_0	MCAN2_RX		SPIMX8_0_SCLK	LCD_DAT4			I2S0_SCLK			DVP_D0
	BT_GPIOA_4	BT_TX_IND													
PB1/BT_PA5	GPIOB_1	UART2_TXD	SPIMX2_0_CSN	I2C1_SDA	PWM0_1	MCAN2_TX		SPIMX8_0_CSN	LCD_DAT5			I2S0_LRCK			DVP_D1
	BT_GPIOA_5	BT_RX_IND													
PB3/BT_PA7	GPIOB_3	UART2_RTS	SPIMX2_0_MISO	I2C0_SDA	PWM0_3			SPIMX8_0_IO1	LCD_DAT7		PDM0_SDA	I2S0_DIN	SPDIF_IN		DVP_D3
	BT_GPIOA_7														
PB4	GPIOB_4	UART3_RXD	SPIS0_SCLK	I2C0_SCK	PWM0_4	MCAN0_RX		SPIMX8_0_IO2	LCD_DAT8						
PB5	GPIOB_5	UART3_TXD	SPIS0_CSN	I2C0_SDA	PWM0_5	MCAN0_TX		SPIMX8_0_IO3	LCD_DAT9						
PB6	GPIOB_6	UART3_CTS	SPIS0_MOSI	I2C1_SCK	PWM0_6	MCAN1_RX	SDIOH0_CLK	SPIMX8_0_SCLK	LCD_DAT10		PDM1_SCK				
PB7	GPIOB_7	UART3_RTS	SPIS0_MISO	I2C1_SDA	PWM0_7	MCAN1_TX	SDIOH0_CMD	SPIMX8_0_CSN	LCD_DAT11		PDM1_SDA				
PB8	GPIOB_8	UART2_RXD	SPIMX2_0_SCLK	I2C0_SCK	PWM0_8	MCAN0_RX	SDIOH0_DAT0	SPIMX8_0_IO4	LCD_DAT12			I2S1_SCLK			
PB9	GPIOB_9	UART2_TXD	SPIMX2_0_CSN	I2C0_SDA	PWM0_9	MCAN0_TX	SDIOH0_DAT1	SPIMX8_0_IO5	LCD_DAT13			I2S1_LRCK			

Confidential

名称	GPIO 功能														
PB10	GPIOB_10	UART2_CTS	SPIMX2_0_MOSI	I2C0_SCK	PWM0_10	MCAN1_RX	SDIOH0_DAT2	SPIMX8_0_IO6	LCD_DAT14	MCU_SWCLK	PDM1_SCK	I2S1_DOUT	SPDIF_OUT		
PB11	GPIOB_11	UART2_RTS	SPIMX2_0_MISO	I2C0_SDA	PWM0_11	MCAN1_TX	SDIOH0_DAT3	SPIMX8_0_IO7	LCD_DAT15	MCU_SWDIO	PDM1_SDA	I2S1_DIN	SPDIF_IN		
PB12	GPIOB_12	UART3_RXD	SPIS0_SCLK	I2C1_SCK	PWM0_12	SPIMX8_0_IO4	SDIOH0_DAT4		LCD_CSX	OSPI_SCLK<XIP>			QSPI1_SCLK<XIP>		
PB13	GPIOB_13	UART3_TXD	SPIS0_CSN	I2C1_SDA	PWM0_13	SPIMX8_0_IO5	SDIOH0_DAT5		LCD_WRX	OSPI_CSN<XIP>			QSPI1_IO0<XIP>		
PB14	GPIOB_14	UART3_CTS	SPIS0_MOSI	I2C0_SCK	PWM0_14	SPIMX8_0_IO6	SDIOH0_DAT6		LCD_RDX	OSPI_DQS<XIP>	PDM1_SCK		QSPI1_IO3<XIP>		
PC1	GPIOC_1	UART2_TXD	SPIMX2_0_CSN	I2C1_SDA	PWM1_1	SPIMX8_0_IO1	SDIOH0_DAT1		LCD_DAT1	OSPI_IO5<XIP>		I2S1_LRCK	QSPI1_CSN<XIP>		
PC0	GPIOC_0	UART2_RXD	SPIMX2_0_SCLK	I2C1_SCK	PWM1_0	SPIMX8_0_IO0	SDIOH0_DAT0		LCD_DAT0	OSPI_IO6<XIP>		I2S1_SCLK	QSPI1_IO1<XIP>		
PB15	GPIOB_15	UART3_RTS	SPIS0_MISO	I2C0_SDA	PWM0_15	SPIMX8_0_IO7	SDIOH0_DAT7		LCD_DCX	OSPI_IO7<XIP>	PDM1_SDA		QSPI1_IO2<XIP>		
PC2	GPIOC_2	UART2_CTS	SPIMX2_0_MOSI	I2C2_SCK	PWM1_2	SPIMX8_0_IO2	SDIOH0_DAT2		LCD_DAT2	OSPI_IO1<XIP>	PDM1_SCK	I2S1_DOUT	SPDIF_OUT		
PC3	GPIOC_3	UART2_RTS	SPIMX2_0_MISO	I2C2_SDA	PWM1_3	SPIMX8_0_IO3	SDIOH0_DAT3		LCD_DAT3	OSPI_IO3<XIP>	PDM1_SDA	I2S1_DIN	SPDIF_IN		
PC4	GPIOC_4	UART3_RXD	SPIS0_SCLK	I2C2_SCK	PWM1_4	SPIMX8_0_SCLK	SDIOH0_CLK		LCD_DAT4	OSPI_IO4<XIP>				DAC_PWMLP/PWM_LCK	
PC5	GPIOC_5	UART3_TXD	SPIS0_CSN	I2C2_SDA	PWM1_5	SPIMX8_0_CSN	SDIOH0_CMD	SPIMX8_1_DQS	LCD_DAT5	OSPI_IO2<XIP>				DAC_PWMLN/PWM_LCK	
PC6	GPIOC_6	UART3_CTS	SPIS0_MOSI	I2C1_SCK	PWM1_6	SPIMX8_0_DQS	SDIOH0_DAT4	SPIMX8_1_IO4	LCD_DAT6	OSPI_IO0<XIP>	PDM1_SCK			DAC_PWMRP/PWM_RCK	
PC7	GPIOC_7	UART3_RTS	SPIS0_MISO	I2C1_SDA	PWM1_7		SDIOH0_DAT5	SPIMX8_1_IO5	LCD_DAT7	OSPI_SCLK#<XIP>	PDM1_SDA			DAC_PWMRN/PWM_RCK	
PD0	GPIOD_0	UART4_RXD	SPIMX2_1_SCLK	I2C2_SCK	PWM1_0	MCAN0_RX	SDIOH0_CLK	SPIMX8_1_SCLK	LCD_DAT0			I2S1_SCLK			DVP_D0
PD1	GPIOD_1	UART4_TXD	SPIMX2_1_CSN	I2C2_SDA	PWM1_1	MCAN0_TX	SDIOH0_CMD	SPIMX8_1_CSN	LCD_DAT1			I2S1_LRCK			DVP_D1
PD2	GPIOD_2	UART4_CTS	SPIMX2_1_MOSI	I2C1_SCK	PWM1_2	MCAN1_RX	SDIOH0_DAT4	SPIMX8_1_IO0	LCD_DAT2		PDM1_SCK	I2S1_DOUT	SPDIF_OUT		DVP_D2
PD3	GPIOD_3	UART4_RTS	SPIMX2_1_MISO	I2C1_SDA	PWM1_3	MCAN1_TX	SDIOH0_DAT5	SPIMX8_1_IO1	LCD_DAT3		PDM1_SDA	I2S1_DIN	SPDIF_IN		DVP_D3
PD4	GPIOD_4	UART5_RXD		I2C2_SCK	PWM1_4	SPIS1_SCLK	SDIOH0_DAT6	SPIMX8_1_IO2	LCD_DAT4					DAC_PWMLP/PWM_LCK	DVP_D4
PD5	GPIOD_5	UART5_TXD		I2C2_SDA	PWM1_5	SPIS1_CSN	SDIOH0_DAT7	SPIMX8_1_IO3	LCD_DAT5					DAC_PWMLN/PWM_LCK	DVP_D5

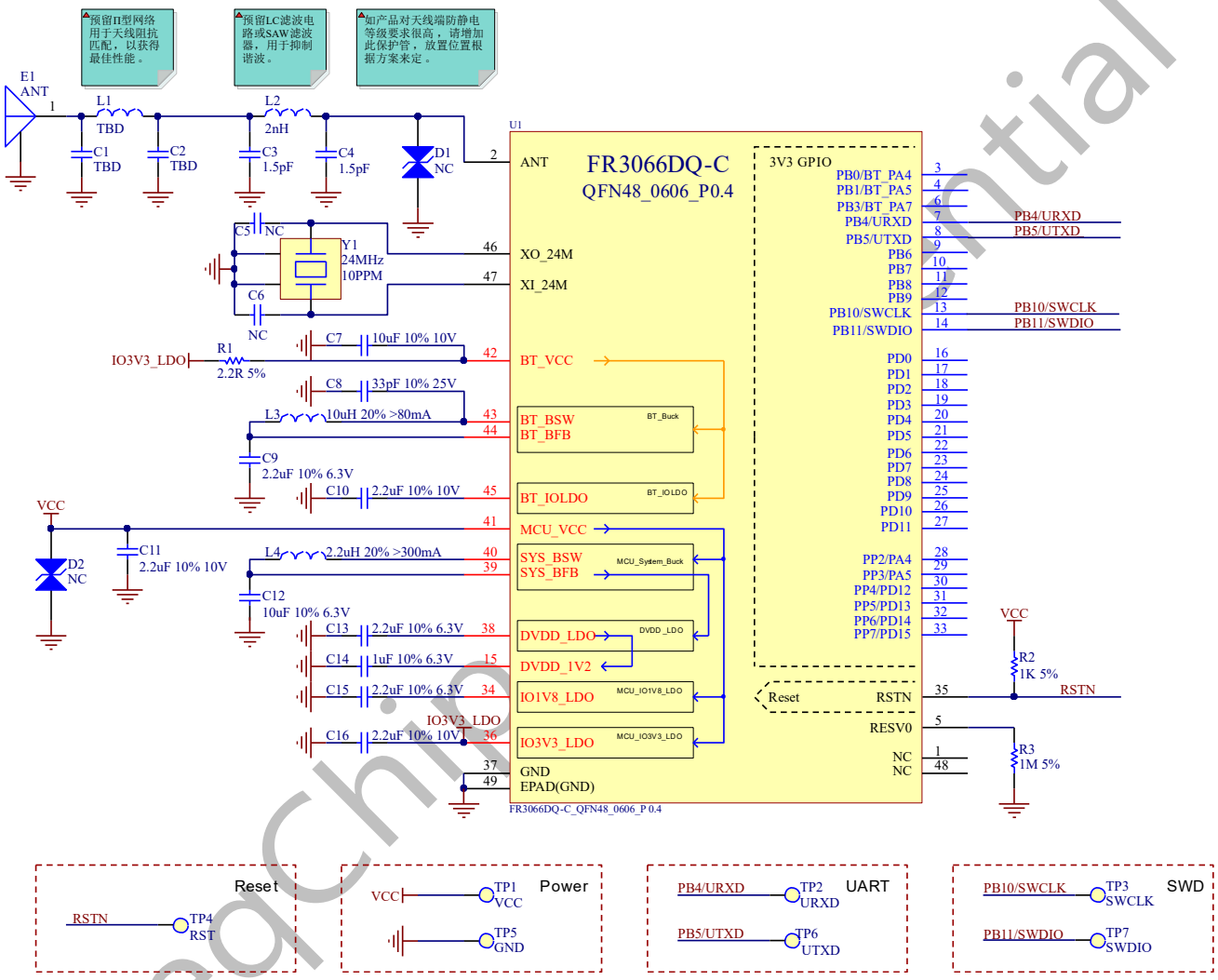
名称	GPIO 功能														
PD6	GPIOD_6	UART5_CTS		I2C2_SCK	PWM1_6	SPIS1_IO0	SDIOH0_DAT0	SPIMX8_1_IO4	LCD_DAT6		PDM1_SCK			DAC_PWMRP/PWM_RCK	DVP_D6
PD7	GPIOD_7	UART5_RTS		I2C2_SDA	PWM1_7	SPIS1_IO1	SDIOH0_DAT1	SPIMX8_1_IO5	LCD_DAT7		PDM1_SDA			DAC_PWMRN/PWM_RCK	DVP_D7
PD8	GPIOD_8	UART4_RXD	SPIMX2_1_SCLK	I2C1_SCK	PWM1_8	SPIS1_IO2	SDIOH0_DAT2	SPIMX8_1_IO6	LCD_CSX			I2S0_SCLK		DAC_PWMCLK	DVP_D8
PD9	GPIOD_9	UART4_TXD	SPIMX2_1_CSN	I2C1_SDA	PWM1_9	SPIS1_IO3	SDIOH0_DAT3	SPIMX8_1_IO7	LCD_WRX			I2S0_LRCK			DVP_D9
PD10	GPIOD_10	UART4_CTS	SPIMX2_1_MOSI	I2C2_SCK	PWM1_10	SPIS1_SCLK	SDIOH0_CLK	SPIMX8_1_DQS	LCD_RDX		PDM1_SCK	I2S0_DOUT	SPDIF_OUT		DVP_D10
PD11	GPIOD_11/ADC	UART4_RTS	SPIMX2_1_MISO	I2C2_SDA	PWM1_11	SPIS1_CSN	SDIOH0_CMD		LCD_DCX		PDM1_SDA	I2S0_DIN	SPDIF_IN	DAC_PWMLP/PWM_LCK	DVP_D11
PD12	GPIOD_12/ADC	UART5_RXD		I2C2_SCK	PWM1_12	MCAN0_RX	SDIOH0_DAT4		LCD_DAT8	USB_DP		I2S1_SCLK		DAC_PWMLN/PWM_LCK	
PD13	GPIOD_13/ADC	UART5_TXD		I2C2_SDA	PWM1_13	MCAN0_TX	SDIOH0_DAT5		LCD_DAT9	USB_DM		I2S1_LRCK		DAC_PWMRP/PWM_RCK	
PD14	GPIOD_14/ADC	UART5_CTS		I2C1_SCK	PWM1_14	MCAN1_RX	SDIOH0_DAT6		LCD_DAT10	USB_DP	PDM1_SCK	I2S1_DOUT	SPDIF_OUT	DAC_PWMRN/PWM_RCK	
PD15	GPIOD_15/ADC	UART5_RTS		I2C1_SDA	PWM1_15	MCAN1_TX	SDIOH0_DAT7		LCD_DAT11	USB_DM	PDM1_SDA	I2S1_DIN	SPDIF_IN	DAC_PWMCLK	
PP4	PMU_IO4/ADC														
PP5	PMU_IO5														
PP6	PMU_IO6														
PP7	PMU_IO7														
PP0	PMU_IO0/ADC/ADKEY0														
PP3	PMU_IO3/ADC														
PP2	PMU_IO2/ADC														
PA4	GPIOA_4	UART1_RXD	SPIS0_SCLK	I2C1_SCK	PWM0_4	MCAN0_RX				USB_DP					
PA5	GPIOA_5	UART1_TXD	SPIS0_CSN	I2C1_SDA	PWM0_5	MCAN0_TX		SPIMX8_0_DQS		USB_DM					
PA6	GPIOA_6	UART1_CTS	SPIS0_MOSI	I2C0_SCK	PWM0_6	MCAN1_RX		SPIMX8_0_SCLK		USB_DP			SPDIF_OUT		
PA7	GPIOA_7	UART1_RTS	SPIS0_MISO	I2C0_SDA	PWM0_7	MCAN1_TX		SPIMX8_0_CSN		USB_DM			SPDIF_IN		

注：PB12 ~ PB15/PC0~PC7 为 1.8V 电压域。

Freqchip Confidential

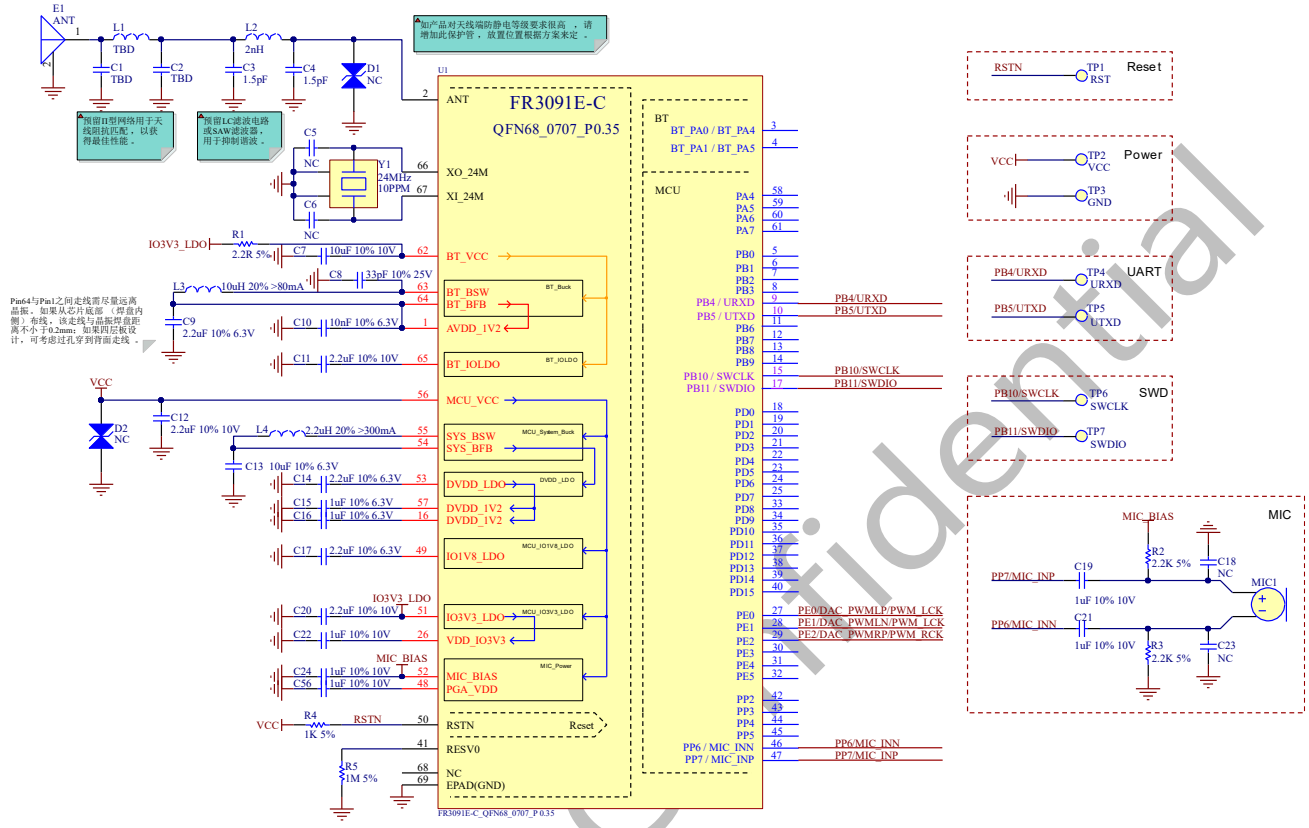
3. 参考设计

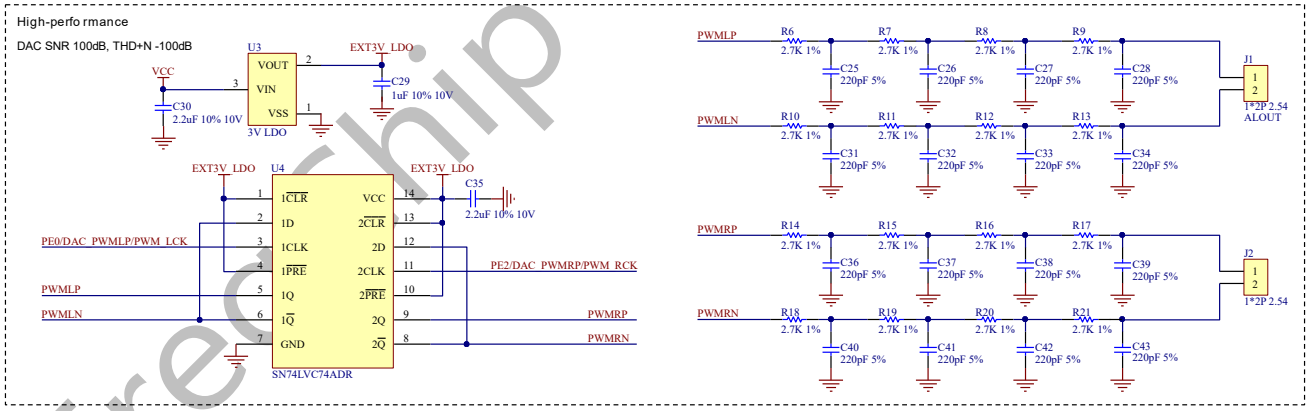
3.1 FR3066DQ-C 参考设计

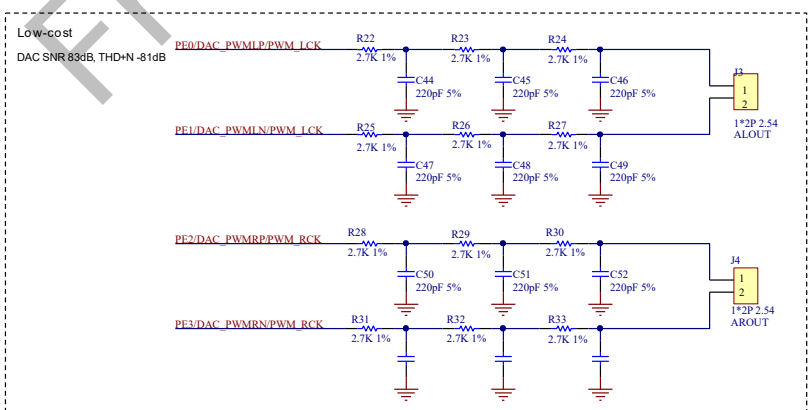


FreqChip Confidential

3.2 FR3091E-C 参考设计

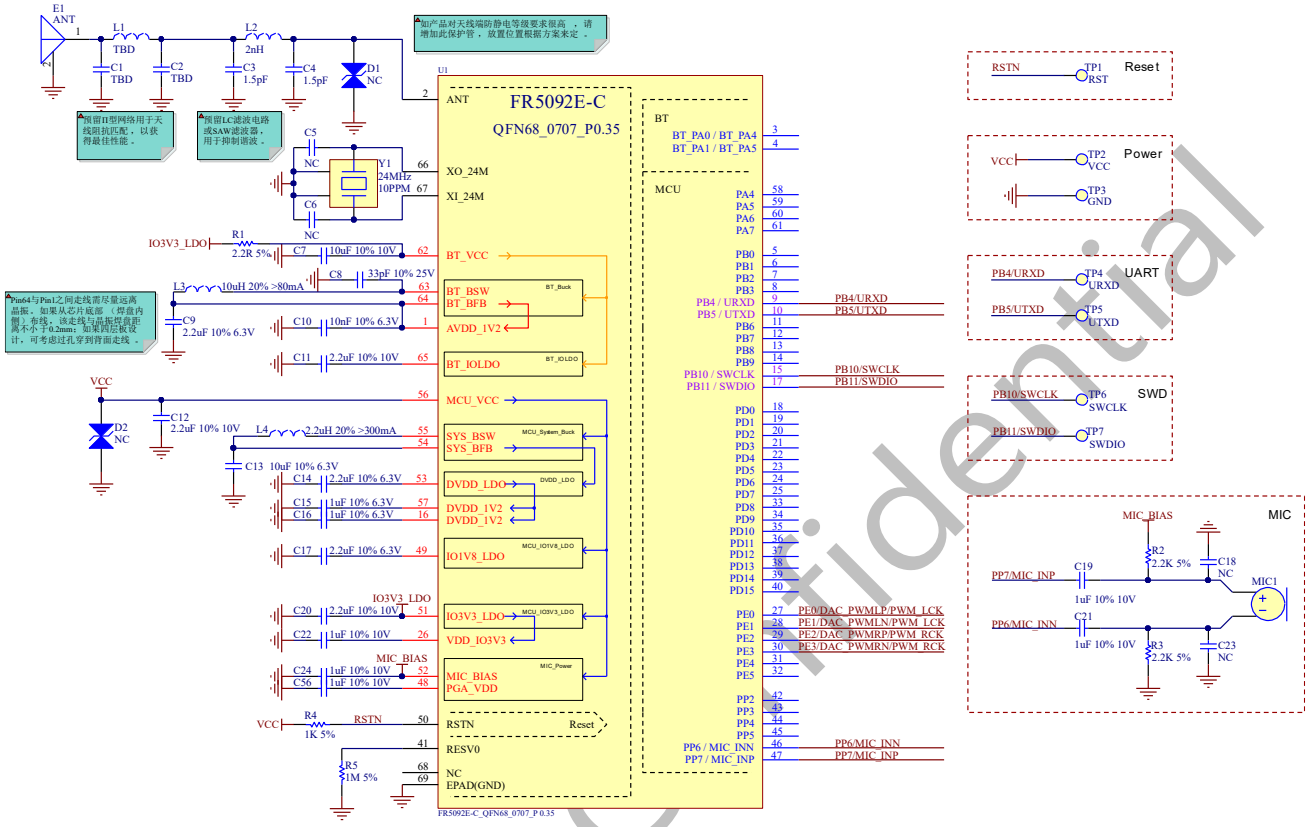






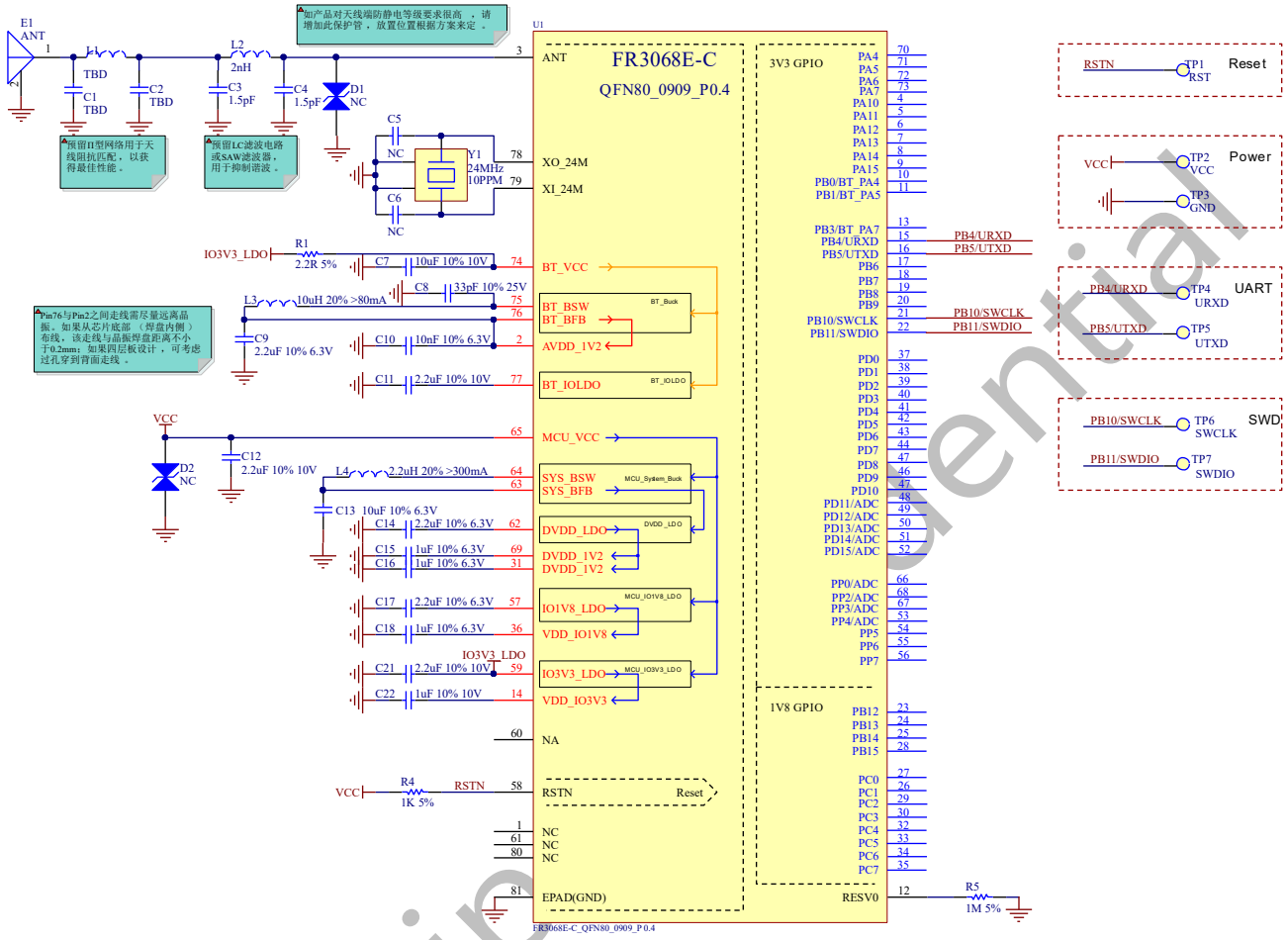
FreqChip Confidential

3.3 FR5092E-C 参考设计



FreqChip Confidential

3.4 FR3068E-C 参考设计

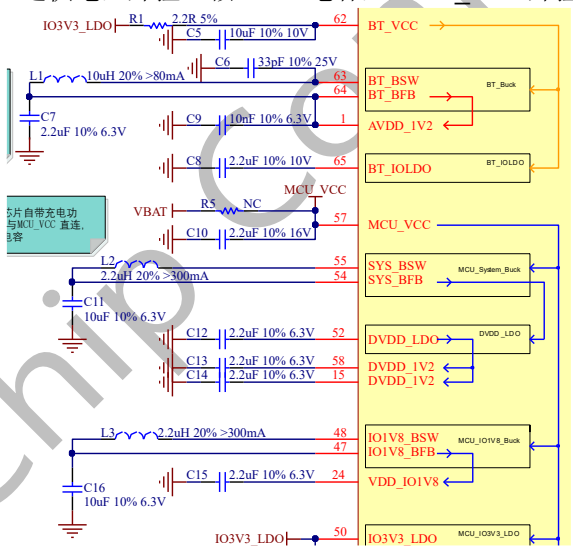


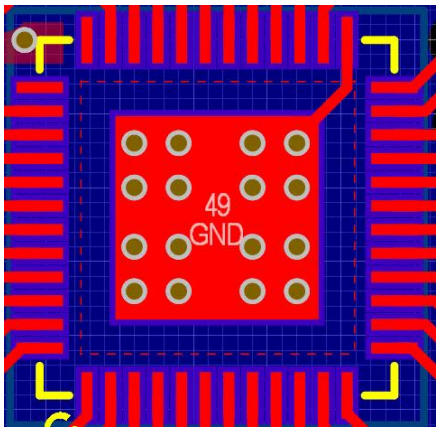
FreqChip Confidential

4. 应用设计注意事项

4.1 电源

- FR30xx-C 供电电压范围为 2.9V~3.6V。
- MCU_VCC、BT_VCC、BT_IOLDO、BT_BFB、SYS_BFB 等电源引脚的滤波电容尽量靠近 IC 相应引脚放置，滤波电容的地必须就近接到 IC 底部的地，确保滤波效果。
- BT_BSW / SYS_BSW 为内部 DC / DC 模块开关信号输出脚，BT_BSW / SYS_BSW 脚电感靠近 IC 引脚放置，滤波电容紧靠电感放置，走线尽量短和宽且先经电容后再进芯片引脚。
- BT 的 Buck 电感值推荐 10uH，额定电流大于 80mA，直流电阻小于 1ohm，选用功率叠层电感或者绕线电感。电感因实际工作电流超过额定电流会导致饱和，造成电感发热、输出纹波、效率变低。
- MCU 的 Buck 电感值推荐 2.2uH，额定电流大于 300mA，直流电阻小于 1ohm，选用功率叠层电感或者绕线电感。电感因实际工作电流超过额定电流会导致饱和，造成电感发热、输出纹波、效率变低。
- 如果使用芯片内部 1.8V/3.3V LDO 给外设供电，建议从 IO1V8_LDO 或 IO3V3_LDO 引脚取电，该 LDO 输出电流 60mA。
- 芯片底部焊盘至少放 9 个地孔连接到地平面，确保接地良好。
- AVDD 1V2 需要 BT_BFB 直连供电，外挂一颗 10nF 电容，且 BT_BFB 外挂一颗 2.2uF。

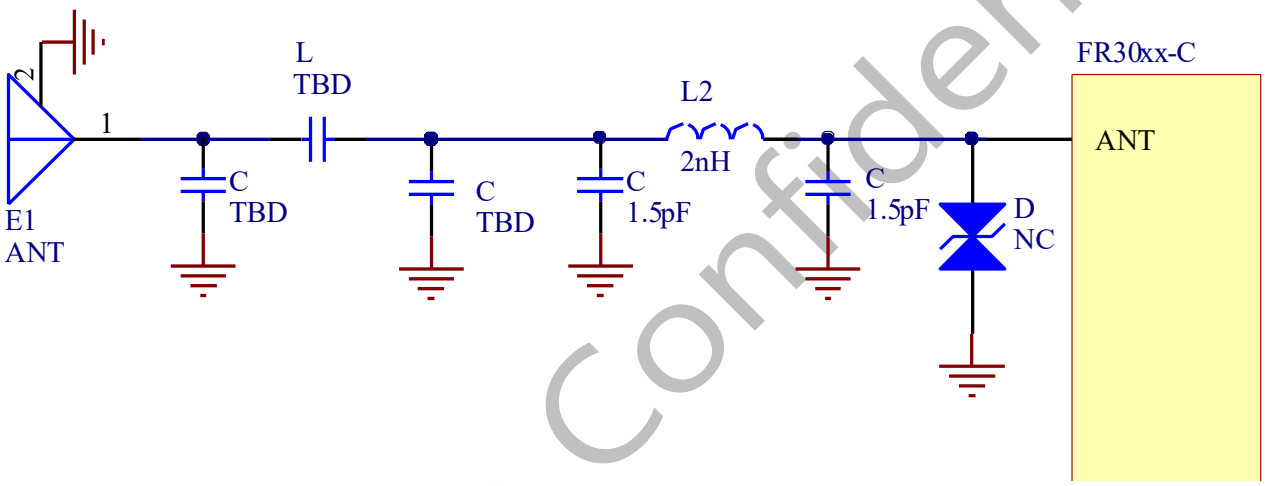


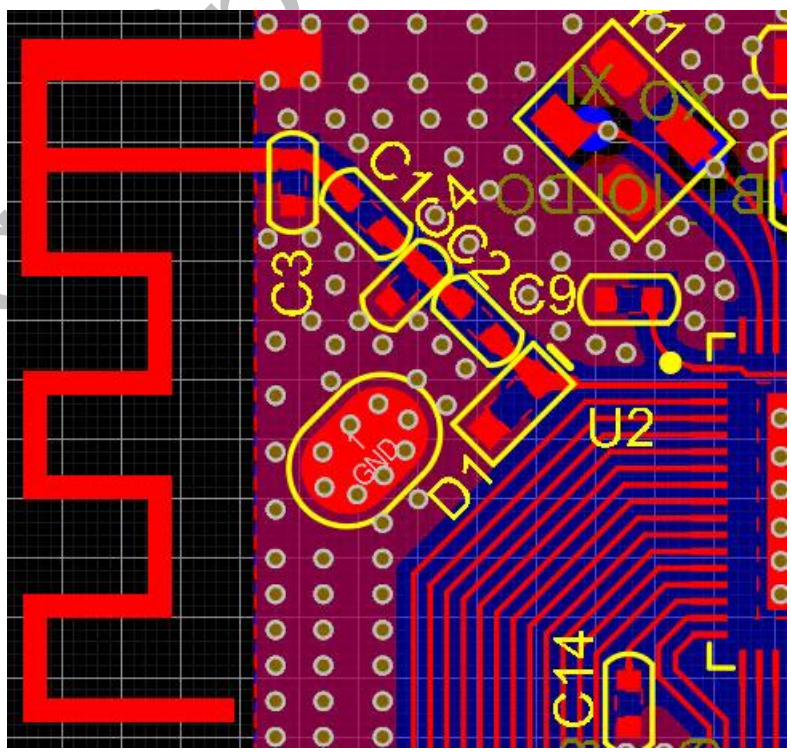


FreqChip Confidential

4.2 射频

- 射频走线需要走 50Ω 阻抗线，走线尽量短和宽，不要有过孔，最好和芯片同层。不能走直角，尽量使用圆弧走线或 135° 角走线，走线宽度保持一致，避免分支走线，周围地孔屏蔽。确保射频走线底层有完整的地平面，并且与芯片底部的 GND 相连。
- 芯片 ANT 引脚端与天线之间务必预留如图所示 π 型谐波抑制电路。
- 预留 π 形匹配电路，靠近芯片 RF 脚放置，用于天线匹配，天线匹配阻抗往 50Ω 方向调，元件参数值根据天线和 PCB 布局实测后确定。
- 天线匹配电路中的电感参数推荐自谐振频率最好大于等于工作频率的三倍、Q 值尽可能大、直流内阻（DRC）尽可能小；电容参数推荐材质优先选择 C0G，高精度，高 Q 值，低 ESR。
- 天线尽量使用倒 F 形。
- 天线必须放置在板边，严禁被 GND 包裹，且正反面需要净空区，远离金属和电池，远离高频噪声源。

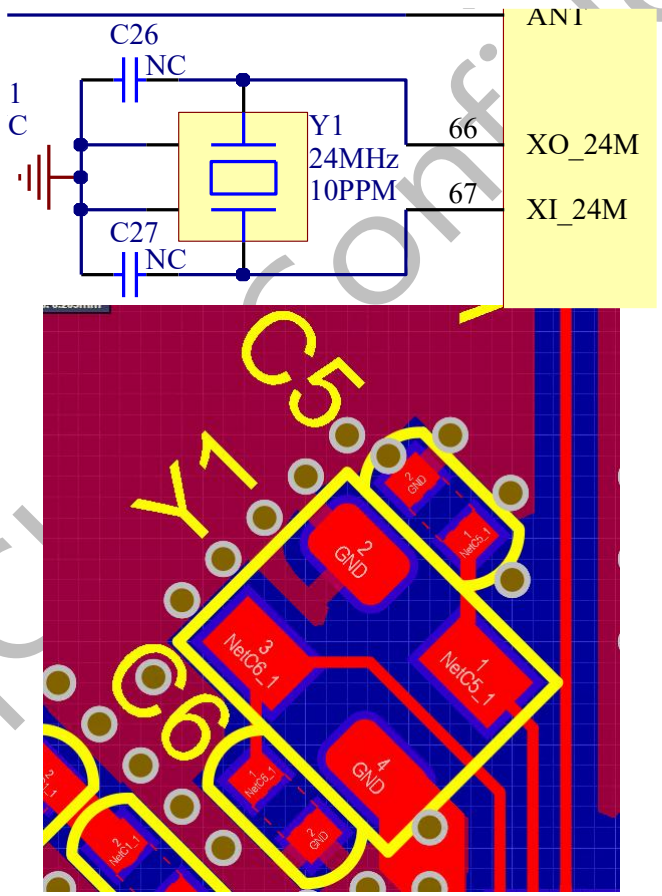




Freqchip Confidential

4.3 晶振

- 晶振走线尽量短，尽量不要有过孔，与 RF 微带线之间用地线隔离。预留的外部负载电容靠近晶振放置，晶振底层尽量不要走其它信号线，晶振周围包地处理。晶振为敏感元件，不得靠近磁感应元件，远离 BSW 脚的电感。
- 24MHz 晶振选型参数推荐使用负载电容 9~12pF、频率公差±10ppm、ESR≤60R、C0≤2pF、封装为 2016 或 3225 的晶振；另外，注意晶振工作温度范围应满足应用需求，优先使用芯片内部的晶振匹配电容。芯片内部晶振负载电容由 PMU 寄存器的 0xEC[3:0]控制，有效取值范围是 0x00~0x0B。9pF 晶振一般配置为 0x01，12pF 晶振一般配置为 0x08，具体配置值以产品实测为准。由于晶振自身频偏过大，工作温度稳定度差等原因，可能会导致 FR30xx-C 性能指标下降，甚至无法工作。推荐选用外壳接地的贴片晶振，性能更加稳定。
- 实际应用中，即使物料不变，由于布局、PCB 板材等差异会产生频偏，因此每个方案量产前都需要取几台样机测量一下频偏，通过调整内部晶振负载电容寄存器的配置值使得 2.4G 上的频偏控制在±25KHz 以内。



FreqChip Confidential

4.4 GPIO

- 所有管脚可以被配置为输入和输出 2 种状态。FR30xx-C 系列 GPIO 管脚分为 3.3V 和 1.8V 两组电压域，3.3V 电压域的 GPIO 管脚电压取决于 IO3V3_LDO 脚电压，1.8V 电压域的 GPIO 管脚电压取决于 IO1V8_LDO 脚电压。GPIO 关键 3.3V 电压域做为输入模式时的门限是，高电平需要 > 0.7 IO3V3_LDO 脚电压，低电平需要< 0.3* IO3V3_LDO 脚电压；GPIO 关键 1.8V 电压域做为输入模式时的门限是，高电平需要 > 0.7 IO1V8_LDO 脚电压，低电平需要< 0.3* IO1V8_LDO 脚电压
- 管脚 PB4、PB5 在系统初始化运行时，内部固化的 rom code 会配置为 uart0 的控制管脚。PB5 会输出内部 uart 打印信息，PB4 会接收外部的打印信息。

4.5 系统设计

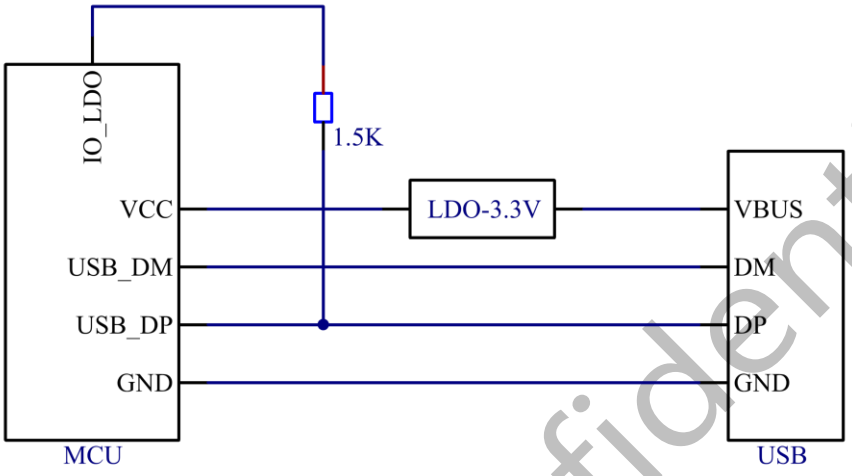
- 每个方案量产前，需要抽取几台样机，根据实际使用的晶振、天线进行频偏和天线阻抗匹配调试。如果不做频偏和天线阻抗匹配调试，会影响射频性能，表现为搜索连接时间长、收发数据不稳定、断线等等。
- 系统中有电机等高功率器件，在 PCB 设计时，务必把蓝牙部分的 GND 回路与高功率器件的 GND 回路分开，然后单点连接到电源地。
- 在设计方案时，建议预留 TP 点：RX（PB4）、TX（PB5）、SWCLK（PB10）、SWDIO（PB11）、VCC、GND 及 Reset；若有空余资源， 建议增加预留 HCI 通讯的 RX（PB0）与 TX（PB1）的 TP 点。

FreqChip Confidential

4.6 应用接口

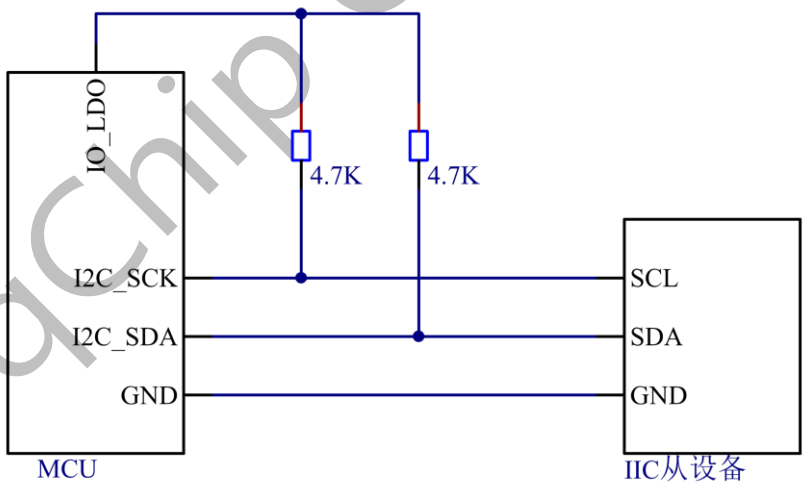
4.6.1 USB

GPIO 作为 USB 功能时，USB_DP 必需加 1.5K 外部上拉电阻接 IO_LDO；



4.6.2 IIC

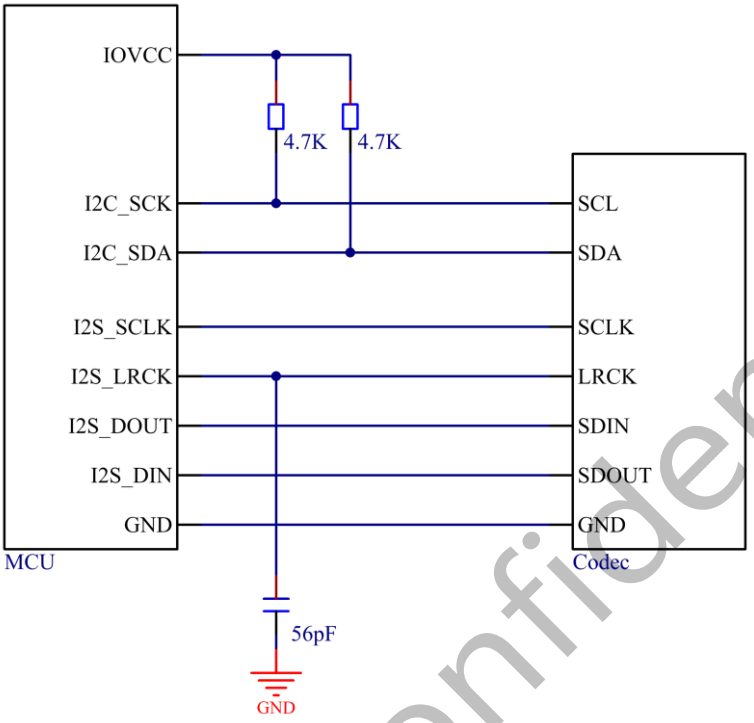
GPIO 作为 I2C 功能时，必需加 4.7K 外部上拉电阻接 IO_LDO；



FreqChip Confidential

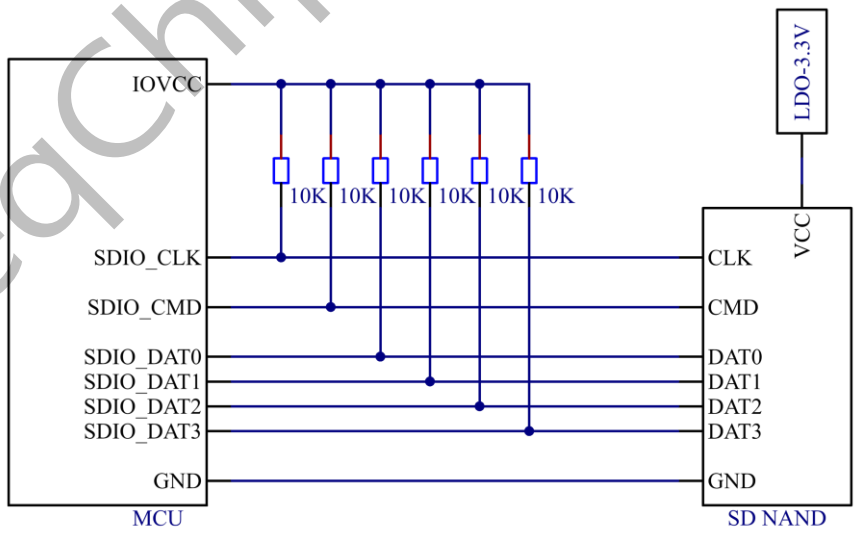
4.6.3 IIS

GPIO 作为 I2S 功能时，I2S LRCK 时钟需并接 56pF 电容到地



4.6.4 SD

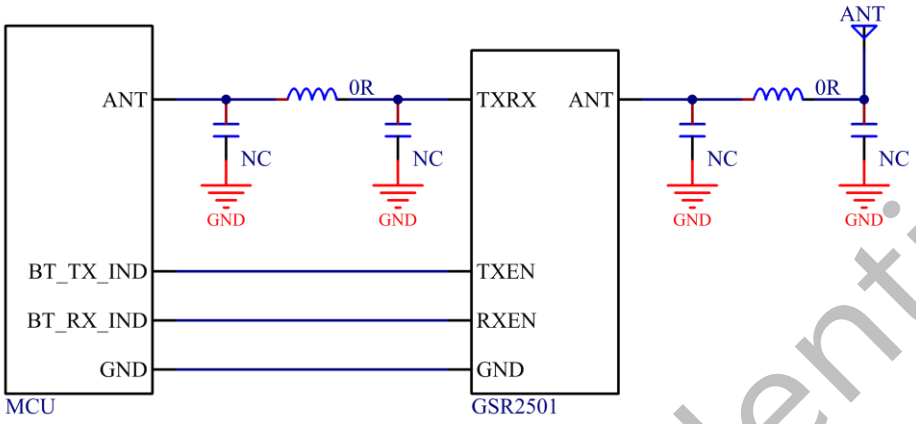
GPIO 与 SD 卡或者 SD Nand 芯片通信时，需增加上拉电阻至通信电平电源，SD 卡或者 SD Nand 芯片采用外置 LDO 独立供电 3.3V；



FreqChip Confidential

4.6.5 外接 2.4G PA

当需要接外部 2.4G PA 时，需要芯片的 ANT 脚预留一个 π 型网络，芯片的 BT_TX 与 BT_RX 使能控制外接 2.4G PA，且外接的 2.4G PA 的 ANT 脚也需要预留一个 π 型网络；

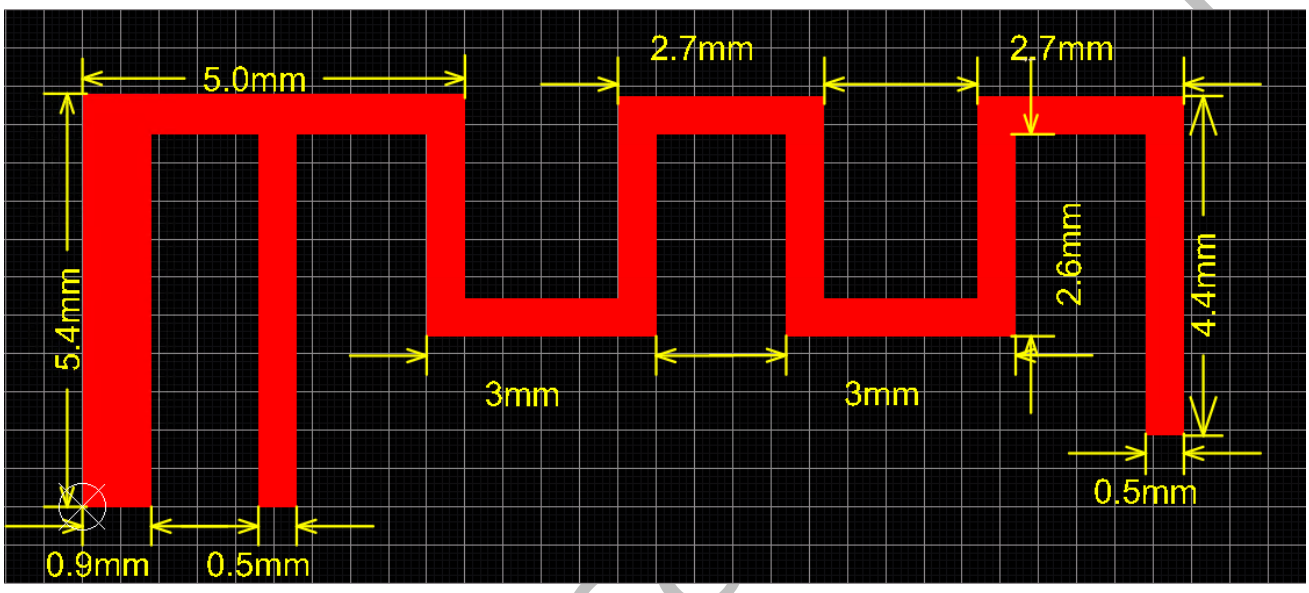


FreqChip Confidential

5. PCB 天线

- PCB 板载天线可满足一般应用需求，但是性能稍差、易受干扰，优点是成本低廉、整机组装方便。对于产品尺寸小、性能要求高的应用，请选用贴片陶瓷天线或外置天线。

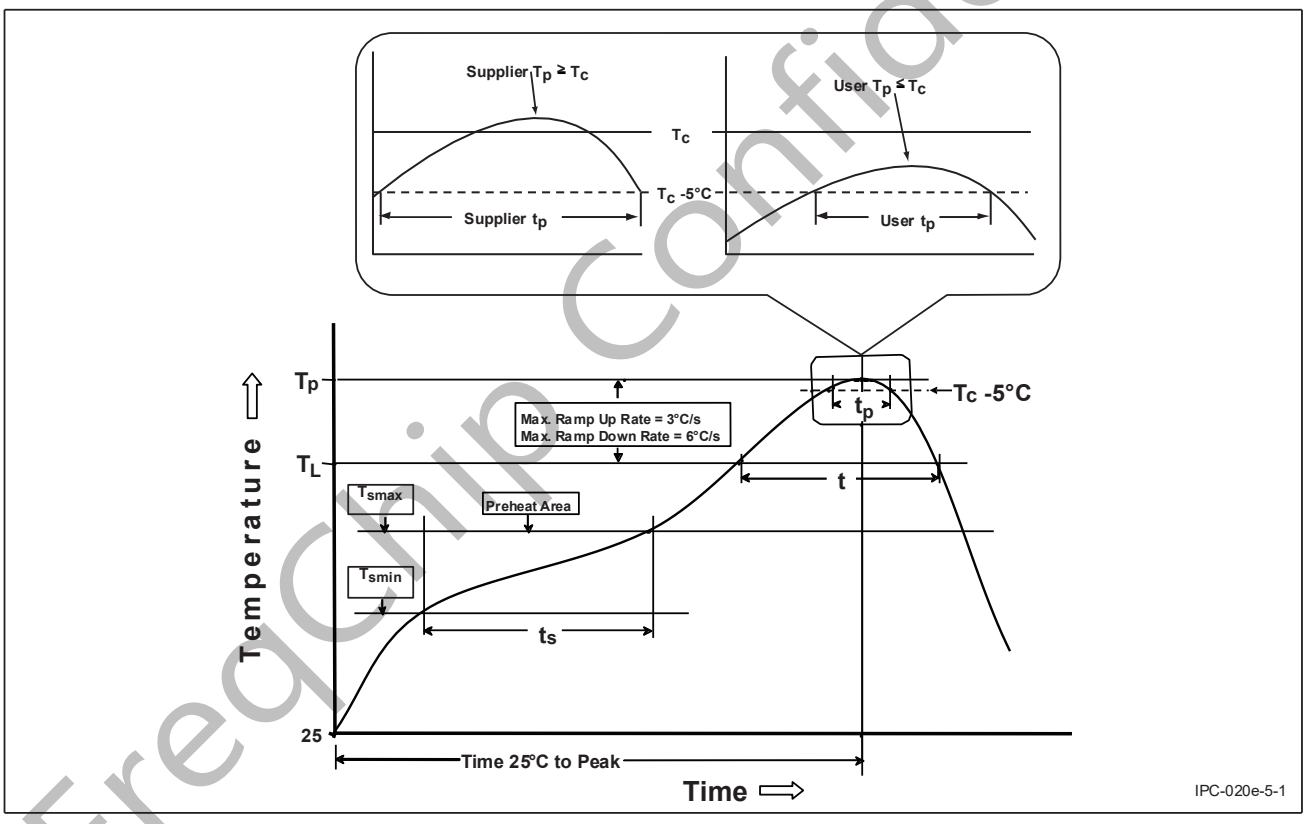
- 倒 F 天线



FreqChip Co

6. 回流焊

Profile Feature	Pb-Free Assembly
Preheat / Soak Temperature Min (T_{smin}) Temperature Max (T_{smax}) Time (t_s) from (T_{smin} to T_{smax})	150 °C 200 °C 60-120 seconds
Ramp-up rate (T_L to T_p)	3 °C / second max.
Liquidous temperature (T_L) Time (t_L) maintained above T_L	217 °C 60-150 seconds
Peak package body temperature (T_p)	For users T_p of not less than 260 °C.
Time (t_p)* within 5 °C of the specified classi- fication temperature (T_c), see Figure 5-1.	30* seconds
Ramp-down rate (T_p to T_L)	6 °C / second max.
Time 25 °C to peak temperature	8 minutes max.
* Tolerance for peak profile temperature (T_p) is defined as a supplier minimum and a user maximum.	



FreqChip Confidential

联系方式 Contact Information

Website: www.freqchip.com
Sales Email: sales@freqchip.com
Phone: +86-21-5027-0080

修订历史 Revision History

Feedback:

Freqchip welcomes feedback on this product and this document. If you have comments or suggestions, please send an email to docs@freqchip.com.

Reversion Number	Reversion Date	Description
V0.0.1	2024.05.24	初稿
V0.0.2	2024.09.13	更新系列部分引脚定义及参考设计
V0.1.0	2024.11.25	更新全系列参考设计，增加 ANT 脚谐波抑制电路及 RSTN 脚增加 1K 上拉电阻
V0.1.1	2024.11.29	更新引脚定义、增加 FR3066DQ-C、FR3091E-C/FR5092E-C、FR3068E-C 引脚复用表
V1.4.0	2024.12.10	增加天线匹配物料选型推荐说明
V1.4.1	2025.01.07	更新 FR3066DQ-C 封装尺寸表
V1.4.2	2025.02.10	更新全系列引脚定义
V1.4.3	2025.05.27	更新 FR3068E-C、FR3091E-C/FR5092E-C 封装图、引脚定义及参考设计
V1.4.4	2025.06.16	新增应用接口说明
V1.4.5	2025.09.11	更新引脚定义，增加 DVP 功能（FR3066D-C 系列不支持该功能）

FreqChip Confidential